Optimal Carbon Tax with Dynamic Skill Heterogeneity

Alexandre Pavlov*

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Abstract

Extending a dynamic Mirrlees model to two goods, one polluting, this paper studies optimal carbon taxation with redistribution. I find that carbon taxes should be nonlinear and age-dependent when labor is distorted and consumption goods differ in their labor complementarity. The social cost of carbon also reflects the effect of pollution on work incentives. When labor and savings wedges are fixed, carbon taxation can partially offset these constraints. In a US calibration, when income taxes can be adjusted, I find that the best linear carbon tax captures 99% of the welfare gains of a fully nonlinear carbon tax. When income wedges are fixed at US-calibrated levels, age-dependent carbon taxes capture 74% of the welfare gain from moving from linear to fully nonlinear carbon taxation. However, these gains are small compared with those of modest income tax reforms, suggesting that greater flexibility in carbon pricing is a poor substitute for income tax reform.

Keywords: Carbon tax, dynamic Mirrlees taxation, second-best taxation
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*Université de Montréal: alexandre.pavlov@umontreal.ca

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1 Introduction

If the ability-to-pay principle motivates progressive income taxation, should it likewise call for progressive carbon taxation? The standard answer is that poor and rich alike should face the same carbon tax with income taxation alone handling redistribution, yet some economists have proposed nonlinear carbon taxes to address unequal emissions or imperfect income taxes (Chancel and Piketty 2015; Stiglitz 2019; Fleurbaey and Kornek 2021; Chancel 2022). How should a government that cares about redistribution design a carbon tax and how well can a nonlinear carbon tax address income tax imperfections?

In this paper, I extend the dynamic Mirrlees income taxation model of Golosov et al. (2016) to a two-good setting where one good generates pollution and characterize the wedges of an optimal allocation. I find that the consumption wedge between the "clean" and the "dirty" good depends on the household's entire skill history, implying a nonlinear and age-dependent carbon tax, when labor is distorted and one good is more complementary with labor. Moreover, each term in the sum of discounted marginal damages in the social cost of carbon is inflated when labor is distorted and pollution discourages work, increasing the carbon tax. When labor is additively separable from both consumption goods and pollution in preferences, the carbon tax is equal to the standard Pigouvian level. These results extend the consumption taxation results of Mirrlees (1976) and carbon taxation results of Jacobs and de Mooij (2015) to a dynamic setting, showing that consumption taxes should be age-varying and that the social cost of carbon should be adjusted to account for lifetime work incentives.

I also study optimal allocations when labor and savings wedges are held at an fixed level, representing the case of a government that does not reform income taxes. I characterize the optimal consumption wedge, which now has terms that may partially offset the labor and savings wedge constraints. The size and direction of the effect depends on how the willingness-to-pay for the dirty good evolves with income and clean good consumption. Whereas the standard dynamic Mirrlees framework considers jointly optimal allocations, this paper is to my knowledge the first to study optimal consumption wedges with constrained labor and savings wedges.

To get an idea of the quantitative importance of the results, I calibrate the model to US data and compare the welfare gains from relaxing restrictions on carbon and income taxes. I find that a linear carbon tax captures nearly all the welfare gain of a fully nonlinear carbon tax when income taxes are optimal. When income taxes are fixed at the calibrated schedules, fully nonlinear carbon taxation improves welfare over the best linear carbon tax, with 74% of the gain captured by age-dependence alone. Nevertheless, this gain is small compared to the gain from fully optimizing income taxes or even enacting modest

income tax reforms. In the calibration, reducing capital income taxes by 0.9% or labor income taxes by 1.2% delivers about the same welfare gain as moving from a linear to an age-dependent carbon tax, suggesting that a more flexible carbon pricing instrument is a poor substitute for income tax reform. If income tax reform is unavailable, age-dependent carbon taxes are a good alternative to full nonlinearity, which can be hard to implement. This finding on dynamic consumption taxation echoes that of Weinzierl (2011) and Farhi and Werning (2013) on income taxation, who likewise show that optimal nonlinear income taxes can be well-approximated with age-dependent income taxation.

My work connects to the literature on optimal income and consumption taxes. Mirrlees (1976) extends the analysis in Mirrlees (1971) to study the optimal taxation of goods, finding that relative leisure complements should be taxed more heavily to discourage their consumption and indirectly encourage work, helping redistribution. Atkinson and Stiglitz (1976) find that, when preferences are weakly separable in consumption and labor, there is no motive for differentiated consumption taxes. In a static setting with non-separable preferences, Jacobs and Boadway (2014) further show that implementing the optimal allocation generally requires nonlinear consumption taxes. Golosov et al. (2003) extend the Atkinson-Stiglitz result to a dynamic setting with weakly separable preferences. This paper extends the Mirrlees (1976) results on consumption taxation to a dynamic setting without assuming weakly separable preferences, showing that the resulting dirty good wedge is not only nonlinear but also history-dependent.

When consumption generates pollution, consumption taxes gain an externality correction role. Cremer et al. (1998) find that the optimal marginal tax rate on the dirty good combines a corrective term and an incentive term. Jacobs and de Mooij (2015) show that the social value of reducing pollution can also depend on how environmental quality interacts with labor supply. When income and consumption taxes are jointly optimal and allowed to be nonlinear, they also show that the labor wedge keeps the standard structure in Mirrlees (1971) and Saez (2001), with no additional externality correction terms. When consumption goods and environmental quality are weakly separable from labor in the utility function, their carbon tax is equal to the social cost of carbon. I extend their analysis to a dynamic setting with persistent pollution and income risk, showing that the externality introduces no additional terms in income wedges.

A parallel literature studies carbon taxation alongside income taxation using the Ramsey approach with linear tax instruments. In a setting where the government uses linear consumption taxes to fund its expenditures, Sandmo (1975) finds that the optimal tax on the dirty good is the sum of an externality correction term and a revenue-raising Ramsey term, whereas the tax formulas for the other goods keep their structure. Bovenberg and de Mooij (1994) and Bovenberg and Goulder (1996) find that the carbon tax should be

lower than the social cost of carbon because it discourages labor supply, even if its revenue finances labor income tax cuts.

Barrage (2020) extends this static analysis to a dynamic setting, putting Golosov et al. (2014) in a Ramsey taxation framework. She finds an optimal formula for the carbon tax, summing production damages and utility damages divided by the marginal cost of public funds. Her calibrated model suggests the optimal carbon tax should be 8% below the first-best carbon tax. Douenne et al. (2026) and Douenne et al. (2025) extend this analysis to a heterogeneous-agent setting and find that the optimal carbon tax is nearly Pigouvian.

When income taxes cannot be adjusted, carbon taxation may help offset distortions elsewhere in the tax system. Fleurbaey and Kornek (2021) show that nonlinear corrective taxes can improve welfare when income taxes are fixed, even when preferences are separable between consumption and labor. In an economy with uninsurable income risk, Wöhrmüller (2025) finds that a uniform carbon tax combined with lump-sum transfers provides additional insurance when labor income taxes are fixed. Barrage (2020) finds that the optimal carbon tax is lower than the first-best level when labor or capital income taxes are fixed. Douenne et al. (2026) find that fixing the labor income tax lowers the carbon tax and fixing the capital income tax raises it. In a setting with uninsurable income risk, Douenne et al. (2025) find that optimal carbon taxation remains close to the social cost of carbon even when income tax rates are fixed at observed levels. These quantitative studies restrict attention to linear carbon taxes, whereas I also consider age-dependent and fully nonlinear wedges.

Section 2 presents the lifecycle economy. Section 3 characterizes the optimal labor, savings, and dirty good wedges. Section 4 characterizes the dirty good wedge when labor and savings wedges are fixed. Section 5 compares the welfare gains from relaxing restrictions on carbon and income taxation, and Section 6 concludes.

2 Environment

The economy is a two-good extension of the dynamic Mirrlees setting of Golosov et al. (2016), where consumption of one of the goods contributes to a persistent pollution stock harming utility. Households face privately observed skill shocks throughout their lives and a social planner allocates consumption and labor to maximize social welfare, providing redistribution and insurance while correcting the externality. I consider a single country and a single cohort to abstract from international and intergenerational equity concerns, so time and age coincide in this model.

2.1 Households and information

There is a unit mass of households living at dates $t = 0, \ldots, T$. A household with skill $\theta_t > 0$ supplies labor l_t and earns labor income $y_t = \theta_t l_t$. Initial skill is drawn from a distribution F_0 with density f_0 and subsequent skills follow a Markov process $F_t(\theta|\theta_{t-1})$ with density f_t. Households are differentiated by their skill history, $\theta^t = (\theta_0, \ldots, \theta_t)$, with Θ^t the set of all possible skill histories up to date t.

Assumption 2.1 (Skill process)**.**
For $t = 1, \ldots, T$, the transition density $f_t(x|s)$ defined on the domain $(0, \infty)$ is differentiable in the previous period's skill. Write $f_{2,t} \equiv \partial f_t / \partial s$. For every compact interval $I \subset (0, \infty)$, there exists $m_{t,I} \in L^1(0, \infty)$ such that

$$\sup_{s \in I} |f_{2,t}(x \mid s)| \leq m_{t,I}(x) \qquad \text{a.e} \quad x > 0$$

Each household consumes a clean good c_t and a dirty good q_t, whose aggregate Q_t generates emissions contributing to a pollution stock S_t affecting utility. Expected lifetime utility is

$$\mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U(c_t, q_t, l_t, S_t) \right]$$

where $\beta \in (0, 1)$ is the discount factor. Both consumption goods increase utility, while labor and pollution decrease it. I allow non-separability in these arguments, meaning consumption and pollution can affect each other's marginal utility as well as that of labor.

The planner observes each household's labor income and consumption of both goods, but neither skill nor labor supply[1]. A high-skill household can therefore earn as much as a low-skill household by working less, such that the planner cannot distinguish them. This creates an information problem constraining the planner's ability to provide insurance and redistribution.

Preferences satisfy a single crossing condition in labor earnings, meaning a high-skill household has a lower marginal cost of providing labor for a given consumption level and pollution stock. Let subscripts on U denote partial derivatives.

Assumption 2.2 (Preferences)**.**
U is twice continuously differentiable, with $U_c, U_q > 0$, $U_l, U_S < 0$, and $U_{cc}, U_{qq}, U_{ll} \leq 0$.

[1]Intuitively, skill may for example be viewed as talent or another unobservable characteristic. Other aspects of skill could be observable but unusable, such as physical characteristics that society might not allow the planner to act upon. Likewise, labor supply may be viewed as effort, which may be difficult to track.

For given labor and pollution levels, preferences over goods are strictly quasi-concave with

$$2U_cU_qU_{cq} - U_c^2U_{qq} - U_q^2U_{cc} > 0$$

I assume strict single crossing in labor earnings, holding (c, q, y, S) fixed,

$$\frac{\partial}{\partial\theta}\left[\frac{U_l(c, q, y/\theta, S)}{\theta U_c(c, q, y/\theta, S)}\right] > 0$$

2.2 Technology and pollution

Output, which equals labor income, can be converted into either consumption good. The clean good is the numéraire and, for simplicity, both goods have unit resource cost. A linear savings technology transfers resources from one period to the next at a constant gross return $R > 0$. In this model, capital does not enter production.

The clean good generates no emissions, while each unit of the dirty good generates a unit of emissions. Let F^t be the unconditional population distribution of skill histories. Aggregate emissions are

$$Q_t = \int_{\Theta^t} q_t(\theta^t) dF^t(\theta^t) \qquad t = 0, \ldots, T \tag{1}$$

Current and past emissions determine the pollution stock.

$$S_t = \mathcal{S}(S_0, \ldots, S_{t-1}, Q_0, \ldots, Q_t) \tag{2}$$

$\mathcal{S}$ is differentiable and non-decreasing in each argument. Emissions can persist and affect pollution accumulation at later dates. This equation can also be rewritten as $S_t = \mathcal{S}(\mathcal{H}_t, Q_t)$, where $\mathcal{H}_t \equiv (S_0, \ldots, S_{t-1}, Q_0, \ldots, Q_{t-1})$. I do not otherwise restrict the functional form of $\mathcal{S}$. Pollution harms utility, but not output.

2.3 Planner's problem and wedges

Following the standard approach in the dynamic Mirrlees literature, I use a direct mechanism to characterize the allocations attainable under these information constraints. In simple terms, this essentially means that the planner asks households for their skill and then allocates consumption and earnings according to their report. The planner commits to an allocation rule at date 0 and there is no hidden saving or trading between households.

So that the planner's allocation is indeed implemented, the mechanism must make truthful reporting by households optimal, meaning it must be incentive-compatible. By the revelation principle, any equilibrium allocation attainable through another mechanism, such as a tax system, can be reproduced by such a direct mechanism.

An allocation specifies non-negative quantities $c_t(\theta^t)$, $q_t(\theta^t)$, and $y_t(\theta^t)$, which are functions of the household's reported skill history θ^t. For given Pareto weights $\alpha(\theta) \geq 0$ on households with initial skill θ_0, the planner maximizes

$$\int_0^\infty \alpha(\theta)\, \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] dF_0(\theta) \tag{3}$$

Feasibility requires that the present value of consumption does not exceed the present value of output plus initial resources, given by initial capital K_0.

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^{T} \frac{c_t(\theta^t) + q_t(\theta^t) - y_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) \leq RK_0 \tag{4}$$

The planner's allocation must be incentive-compatible, meaning that households must weakly prefer to report their true skill. Let $\sigma_t(\theta^t)$ be the report of a household θ^t and let $\sigma^t(\theta^t) = (\sigma_0(\theta^0), \ldots, \sigma_t(\theta^t))$ be its reporting history up to period t. This translates to the following constraint for every household and every reporting strategy.

$$\begin{aligned} &\mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \\ &\geq \mathbb{E}_0 \left[\sum_{t=0}^{T} \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \end{aligned} \tag{5}$$

I focus on allocations and the marginal distortions, or wedges, they imply for the optimality conditions on labor supply, savings, and consumption decisions. However, tax implementation is beyond the scope of this paper, since it's non-unique and matching household and planner optimality conditions is insufficient to establish implementation. See Golosov et al. (2007) and Kocherlakota (2010) for a further discussion on direct mechanisms and the distinction between its allocations and tax implementation.

The planner's problem is to choose an allocation that maximizes the social welfare function (3) subject to the resource constraint (4), incentive compatibility constraints (5), and pollution accumulation constraints (2). I write $U_{c,t}$, $U_{q,t}$, and $U_{l,t}$ as the marginal utilities of the clean good, dirty good, and labor. Allocations are characterized by wedges on labor

supply, savings, and the consumption mix.

$$1-\tau_t^y(\theta^t) \equiv -\frac{U_{l,t}}{\theta_t U_{c,t}} \qquad 1-\tau_t^s(\theta^t) \equiv \frac{1}{\beta R}\frac{U_{c,t}}{\mathbb{E}_t[U_{c,t+1}]} \qquad 1+\tau_t^q(\theta^t) \equiv \frac{U_{q,t}}{U_{c,t}}$$

Next section, I characterize optimal wedges as a function of elasticities and the skill process, first allowing the planner to freely choose any incentive-feasible allocation.

3 Full optimum

3.1 Recursive problem

Global incentive compatibility compares truthful reporting with every possible reporting strategy, so the constraint set is intractable. I follow the first-order approach of Kapička (2013) and Golosov et al. (2016), solving the dual problem of minimizing the present value of resource costs required to deliver a given utility with global constraints replaced by local envelope conditions.

Write $\theta^t = (\theta^{t-1}, \theta)$ and suppose that θ^{t-1} is fixed. Let $u(\theta)$ be the remaining lifetime utility after the current skill draw, where $w(\theta)$ is its continuation value.

$$u(\theta) = U\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S_t\right) + \beta w(\theta) \tag{6}$$

Let $\dot{u}(\theta)$ represent how remaining lifetime utility changes with the current skill draw. Incentive compatibility implies the following local envelope condition, where $w_2(\theta)$ is how the continuation value changes with current skill.

$$\dot{u}(\theta) = -U_l\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S_t\right)\frac{y(\theta)}{\theta^2} + \beta w_2(\theta) \tag{7}$$

These equations introduce two continuation promises that tomorrow's allocation must deliver, namely that expected utility must be $w(\theta)$ and its sensitivity with respect to current skill must be $w_2(\theta)$. The second promise is important because households with different skills can make the same report, receiving the same continuation allocation. However, under persistent shocks, they may face different future skill distributions so that a nearby household contemplating deviating towards this report may end up receiving a different expected utility.

For $t \geq 1$, the planner's current allocation must respect the promise-keeping and threat-keeping constraints inherited from the previous period.

$$\hat{w}_t(\theta^{t-1}) = \int_0^\infty u(\theta) f_t(\theta|\theta_{t-1}) d\theta \tag{8}$$

$$\hat{w}_{2,t}(\theta^{t-1}) = \int_0^\infty u(\theta) f_{2,t}(\theta|\theta_{t-1}) d\theta \tag{9}$$

Let $V_t(\hat{w}_t(\cdot), \hat{w}_{2,t}(\cdot), \mathcal{H}_t)$ be the minimized resource cost of delivering the promised utility and sensitivity to the whole population. The two promise functions specify the planner's commitments at every preceding history. Since households share the same pollution stock, continuation costs must be minimized jointly. The planner's cost minimization problem is

$$V_t(\hat{w}_t(\cdot), \hat{w}_{2,t}(\cdot), \mathcal{H}_t) = \min_{c_t, q_t, y_t, u_t, w_t, w_{2,t}, Q_t, S_t} \left\{ \int [c_t + q_t - y_t] dF^t + \frac{1}{R} V_{t+1}(w_t(\cdot), w_{2,t}(\cdot), \mathcal{H}_{t+1}) \right\}$$

The planner's problem is the above subject to the remaining lifetime utility and its derivative (6)-(7), promise-keeping and threat-keeping (8)-(9), aggregate dirty good consumption (1), and pollution accumulation (2). The household constraints hold at each preceding history, whereas emissions and pollution are common to the population.

At date 0, there are no inherited promises. The problem $V_0(\hat{w}_0, \mathcal{H}_0)$ uses the initial skill density f_0 and imposes the following weighted expectated utility target.

$$\hat{w}_0 = \int_0^\infty \alpha(\theta) u_0(\theta) dF_0(\theta)$$

The relaxed welfare problem selects the largest $\hat{w}_0$ such that $V_0(\hat{w}_0, \mathcal{H}_0) \leq RK_0$. Write $\bar{\alpha}_0(\theta) = \alpha(\theta)$ and $\bar{\alpha}_t(\theta) = 1$ for $t \geq 1$ to distinguish the initial welfare weights from later promise-keeping constraints. At date T, the continuation promises are fixed at $w_T(\cdot) = w_{2,T}(\cdot) = 0$ and $V_{T+1} = 0$. Since no period after T exists, the planner does not consider subsequent pollution damages.

In the next subsection, I characterize the interior optima of this relaxed problem, assuming continuously differentiable allocation rules and a differentiable continuation value with integrable bounds allowing differentiation under integrals. The density is always positive in the support. The optimality conditions are necessary but not sufficient for global incentive compatibility, which is instead verified numerically.

3.2 Labor, savings, and dirty good wedges

The externality adds no correction term to the labor and savings optimality conditions, though the addition of the second good adds an Edgeworth substitutability term to the labor wedge. Before deriving the wedges implied by the optimal allocation, I define the following elasticities that characterize them.

$$\epsilon_t(\theta) = \frac{U_{ll,t} l_t}{U_{l,t}} \quad \sigma_t(\theta) = -\frac{U_{cc,t} c_t}{U_{c,t}} \quad \sigma_{cq,t}(\theta) = -\frac{U_{cq,t} q_t}{U_{c,t}} \quad \gamma_{c,t}(\theta) = \frac{U_{cl,t} l_t}{U_{c,t}} \quad \gamma_{q,t}(\theta) = \frac{U_{ql,t} l_t}{U_{q,t}}$$

The first two are the inverses of the elasticity of labor supply and of the elasticity of intertemporal substitution. The third is minus the elasticity of the marginal utility of the numéraire with respect to dirty good consumption and is positive when the goods are Edgeworth substitutes. The last two are the complementarity between labor and each of the goods. The strict single crossing condition of Assumption 2.2 implies $1 + \epsilon_t - \gamma_{c,t} > 0$.

Proposition 3.1 (Optimal labor wedge)**.**
Under the boundary and convergence conditions in Appendix A.2, the labor wedge satisfies the following optimality condition for $t \geq 1$

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

where

$$\begin{aligned}
A_t(\theta) &= 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \\
B_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right) d\tilde{x}\right] \\
&\qquad (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x) dx}{\theta f_t(\theta)}
\end{aligned}$$

Proof. See Appendix A.2. □

This is the standard labor wedge formula from Golosov et al. (2016), except that C_t now includes the term $\sigma_{cq,t} \dot{q}_t / q_t$. When the goods are Edgeworth substitutes, higher dirty good

consumption lowers the marginal utility of the numéraire, holding other arguments fixed. If dirty good consumption increases with skill, this increases the exponent term, affecting the gains from redistribution and insurance.

Proposition 3.2 (Optimal savings wedge)**.**
Let $\widehat{U}_{c,t} \equiv \frac{U_{c,t}}{1-\chi_t}$ and $\chi_t \equiv \frac{\gamma_{c,t}}{1+\epsilon_t-\gamma_{c,t}} \frac{\tau_t^y}{1-\tau_t^y}$. For $t < T$, suppose $1 - \chi_t > 0$ at the current history and $1 - \chi_{t+1} > 0$ at every next-period history. Savings then satisfy the following optimality condition

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

Proof. See Appendix A.3. □

This is the generalized inverse Euler condition, expressed as a reweighted inverse Euler condition similar to Hellwig (2021). When preferences are non-separable between labor and consumption, the cost of delivering utility is weighted by $1 - \chi_t$. When the numéraire is a labor complement, consumption makes work less painful and shrinks the information rent that truth-telling requires. Neither the second good nor the externality changes the structure of savings wedges.

Proposition 3.3 (Optimal dirty good wedge)**.**
The dirty good wedge between the clean and dirty good satisfies

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1 + \tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}$$

Proof. See Appendix A.4. □

This optimality condition balances the cost of distorting consumption choices on the left-hand side against the benefit of relaxing incentive constraints on the right-hand side, after accounting for pollution. To get an intuition for this condition, consider a utility-compensated reduction in dirty good consumption and see what happens each side.

Each unit of dirty good taken away requires $1+\tau^q$ units of the clean good as compensation. However, it also saves a unit of this dirty good and avoids emissions, valued at the social cost of carbon. The left-hand side is therefore the marginal cost of the distortion net of the saved dirty good and avoided damages, per unit of the numéraire used as compensation.

If $\tau^y > 0$ and $\gamma_{c,t} \neq \gamma_{q,t}$, meaning there is motive for redistribution and the consumption mix affects labor disutility, the distortion also has an incentive-provision benefit. When $\gamma_{c,t} > \gamma_{q,t}$, meaning the dirty good is a relative labor substitute, a high-skill household

considering to misreport as a low-skill household to be able to work less for the same income ends up valuing the dirty good more than the intended household. When the planner takes away some dirty good consumption and replaces it with more clean good consumption in this low-skill household allocation, the high-skill household may prefer to stay with its intended allocation, relaxing the incentive constraints limiting redistribution and insurance. The planner can use this to increase the earnings it demands from households, with appropriate consumption compensation, whose cost is captured by the labor wedge on the right-hand side. $\gamma_{c,t} - \gamma_{q,t}$ measures how much changing the consumption mix discourages imitation, while $1 + \epsilon - \gamma$ measures how much increasing required earnings encourages it. The ratio reflects how much the planner can ask of households while preserving incentive compatibility. At the optimum, this marginal benefit is equalized with the marginal cost of distorting consumption choices.

When $\tau^y = 0$ or $\gamma_{c,t} = \gamma_{q,t}$, the dirty good wedge is equal to the incentive-adjusted social cost of carbon. Indeed, the right-hand side is then zero and we can rearrange the equation to get $\tau^q = \text{SCC}_t$. There is then no additional incentive correction to the dirty good wedge. As I will show in the next subsection, the social cost of carbon can still differ from the standard Pigouvian rule because pollution may affect work incentives.

This is the dynamic analogue to the dirty good wedge derived in Jacobs and de Mooij (2015). The intratemporal tradeoff keeps the same form, but the labor wedge term causes the right-hand side of the equation to reflect the incentive benefits on both redistribution across initial types and insurance against subsequent skill shocks. As a consequence, the dirty good wedge is not only nonlinear, but also history-dependent. This result extends the dynamic results of Golosov et al. (2003), relaxing the weak separability assumption behind their Atkinson-Stiglitz result.

3.3 Social cost of carbon

Define $\xi_t \equiv \frac{U_{Sl,t} l_t}{U_{S,t}}$ as the complementarity between environmental quality and labor supply. $\xi_t - \gamma_{c,t}$ measures this complementarity relative to that of the numéraire. When it is positive, households working more are willing to give up more clean good consumption for a marginal improvement in environmental quality. When the planner wants to redistribute income and provide insurance against income risk, the social cost of carbon values the damages caused by an additional unit of emissions, adjusted for their effect on incentives.

In the formulas below, $\partial S_{t+j}/\partial Q_t$ is the total effect of current emissions on current and future pollution stocks, including their effects through intermediate pollution stocks. The integrals are taken over the unconditional population distribution F^{t+j} of complete skill histories at each date.

Proposition 3.4 (Social cost of carbon)**.**
The incentive-adjusted social cost of carbon is

$$\mathrm{SCC}_t = -\sum_{j=0}^{T-t} \frac{1}{R^j} \int \left(1 + \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \frac{\tau^y_{t+j}}{1 - \tau^y_{t+j}}\right) \frac{U_{S,t+j}}{U_{c,t+j}} \frac{\partial S_{t+j}}{\partial Q_t} \, dF^{t+j}$$

The social cost of carbon reduces to the standard Pigouvian rule when, at every date and history affected by current emissions, either $\tau^y_{t+j} = 0$ or $\xi_{t+j} = \gamma_{c,t+j}$.

$$\mathrm{SCC}^P_t = -\sum_{j=0}^{T-t} \frac{1}{R^j} \int \frac{U_{S,t+j}}{U_{c,t+j}} \frac{\partial S_{t+j}}{\partial Q_t} \, dF^{t+j}$$

Proof. See Appendix A.5. □

This adjusted social cost of carbon measures the direct effect of pollution on utility, as well as its indirect effect on labor provision incentives. To understand what the bracket term means, suppose the planner considers allowing a marginal increase in emissions, holding utility constant. Since each unit of pollution causes $-U_S/U_c$ in damages in units of the numéraire, the planner must compensate households with $-U_S/U_c$ more of the numéraire to keep them indifferent. Suppose environmental quality is a relative labor complement, so that $\xi - \gamma_c > 0$. Consider a high-skill household contemplating getting the allocation intended for a low-skill household. Because it would work less to produce the same income, it needs less clean consumption to compensate for the environmental harm. Therefore, the compensation it receives at its current labor supply is more than it needs to offset its valuation of the environmental harm, increasing its incentive to misreport its skill.

This effect can be offset by asking the low-skill household to generate less income and adjusting clean consumption to maintain indifference. Each unit of forgone labor income allows clean consumption to fall by $1 - \tau^y$, leaving a net resource loss of τ^y. Compensating the direct environmental harm therefore leaves an additional resource cost associated with restoring incentive compatibility. $\xi - \gamma_c$ measures how strongly pollution increases deviation incentives, while $1 + \epsilon - \gamma_c$ measures how a reduction in required labor income offsets this increase. The ratio determines how demanded labor income must be adjusted to maintain incentive compatibility. When $\tau^y > 0$ and $1 + \epsilon - \gamma_c > 0$, this raises the valuation of pollution above marginal damages.

When environmental quality is instead a relative labor substitute, the argument runs in reverse. This adjusted formula extends that of Jacobs and de Mooij (2015) to a dynamic setting. Current emissions affect incentives not only in the current period, but also every future period when they are persistent. The social cost of carbon therefore depends on

the entire life path of the labor wedge, taking into account not only intratemporal but also intertemporal redistribution.

4 Constrained optimum

The results so far let the government reach any incentive-compatible allocation, but actual carbon taxes may be set inside imperfect tax systems where the income tax is not the one a dynamic Mirrlees optimum might prescribe. To model this restriction, I consider the planner problem in which labor and savings wedges are held at exogenous schedules. That is, of the incentive-feasible allocations, the planner may only offer those producing the required wedges.

Keeping income wedges fixed rules out using carbon revenue to cut marginal income tax rates, as in the revenue-neutral swaps of British Columbia and Sweden, or to fund income-contingent transfers such as France's Chèque énergie. Since the planner chooses the best possible allocation under these restrictions, it implicitly recycles any carbon revenue optimally.

4.1 Fixed income wedges

Let the labor wedge be held at a schedule $\bar{\tau}_t^y(\theta) < 1$ with Lagrange multiplier $\eta_t(\theta) f_t(\theta|\theta_-)$.

$$\Gamma_t(\theta) \equiv -\frac{U_l\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{\theta\, U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} - (1 - \bar{\tau}_t^y(\theta)) = 0 \tag{10}$$

$\Gamma_t(\theta)$ is the distance between the labor wedge the planner would want and the fixed schedule. Moreover, the savings wedge is held at $\bar{\tau}_t^s(\theta_t) < 1$ with Lagrange multiplier $\vartheta_{t+1}(\theta^t) f_t(\theta|\theta_-)$.

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}_t^s(\theta_t))\, \beta R\ \mathbb{E}_t\Big[U_{c,t+1}(\theta^{t+1})\Big] \tag{11}$$

Because the savings-wedge constraint involves current and future consumption, the date-t problem cannot enforce it alone. Given a marginal utility of current consumption, the constraint requires tomorrow's marginal utility to average some value M. Today's consumption is likewise constrained by the requirement $\hat{M}$ inherited from yesterday's problem. This new state is needed because two continuation plans delivering the same promised utility and marginal continuation value can still differ in their expected future marginal utility, even if the savings-wedge constraint only allows one value for the expected future marginal utility. The planner's problem is that of Section 3 with wedge constraints

(10) and (11), M in the continuation value, and state $\hat{M}$.

The labor-wedge constraint adds a term to every first-order condition while the savings-wedge constraint adds two. The first term comes from $\hat{M}$ inherited from the date-$(t-1)$ problem and the second term from M given to the date-$(t+1)$ problem. These two terms can be collected into a single shadow price of the intertemporal wedge constraint:

$$\eta_t^s(\theta^t) \equiv \left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1-\bar{\tau}_t^s)\,\beta R^2}\right] U_{c,t}(\theta^t) \tag{12}$$

Hence, η_t prices the labor-wedge constraint and η_t^s prices the savings-wedge constraint. Both enter the first-order conditions through $\zeta_t \equiv \eta_t(1-\bar{\tau}_t^y) - \eta_t^s$ and η_t enters once more on its own.

To simplify notation and improve interpretability, I define $\mathcal{I}_{c,t}^q = \partial/\partial c_t \ln(U_{q,t}/U_{c,t})$ as the change in the household's willingness to give up a unit of the clean good for a unit of the dirty good as it consumes more of the clean good, and $\mathcal{I}_{y,t}^q = \partial/\partial y_t \ln(U_{q,t}/U_{c,t})$ as its counterpart for labor income. $\mathcal{I}_{c,t}^S = \partial/\partial c_t \ln(-U_{S,t}/U_{c,t})$ and $\mathcal{I}_{y,t}^S = \partial/\partial y_t \ln(-U_{S,t}/U_{c,t})$ are defined similarly for environmental quality.

Proposition 4.1 (Constrained-optimal carbon price)**.**
Define $\tau_t^{y,*}$ as the desired labor wedge. The constrained-optimal dirty good wedge satisfies

$$\frac{\tau_t^q - \mathrm{SCC}_t}{1+\tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1+\epsilon_t - \gamma_{c,t}} \frac{\tau_t^{y,*}}{1-\tau_t^{y,*}} - \eta_t\,\mathcal{I}_{y,t}^q - \zeta_t\,\mathcal{I}_{c,t}^q$$

The social cost of carbon keeps the usual form, with the fiscal adjustment term replaced by

$$1 + \frac{\xi_{t+j} - \gamma_{c,t+j}}{1+\epsilon_{t+j} - \gamma_{c,t+j}} \frac{\tau_{t+j}^{y,*}}{1-\tau_{t+j}^{y,*}} + \eta_{t+j}\,\mathcal{I}_{y,t+j}^S + \zeta_{t+j}\,\mathcal{I}_{c,t+j}^S$$

Proof. See Appendix B.1. □

The first term of the dirty good wedge has the same form as that in the full optimum, with the labor wedge being the one the planner wants but cannot necessarily achieve.

The second term reflects the shadow cost of the labor-wedge constraint. A positive η_t means that lowering the labor wedge would reduce the planner's resource cost of delivering an allocation; the planner would like to lower the labor wedge but cannot. For brevity, I will call this "overdistortion". Likewise, I will call the opposite case "underdistortion". $\mathcal{I}_{y,t}^q$

measures how willingness to pay for the dirty good changes with labor income. Suppose $\eta_t > 0$ and the dirty good is the relative labor substitute, $\mathcal{I}^q_{y,t} < 0$. Then, $-\eta_t \mathcal{I}^q_{y,t} > 0$, pulling up the dirty good wedge. Intuitively, shifting expenditure from the dirty good to the clean good makes work less painful, so a higher carbon price encourages work and undoes part of the labor distortion.

The third term captures how the consumption mix affects the two wedge constraints through the marginal utility of the clean good. ζ_t is the net shadow price of a proportional change in this marginal utility, combining the labor-wedge and savings-wedge constraint multipliers. $\mathcal{I}^q_{c,t}$ measures how willingness to pay for the dirty good changes with clean-good consumption and is positive when the dirty good is normal. Set the savings multiplier aside and suppose $\eta_t > 0$, so that $\zeta_t = \eta_t(1 - \bar{\tau}^y_t) > 0$. If the dirty good is normal, $\mathcal{I}^q_{c,t} > 0$, then $-\zeta_t \mathcal{I}^q_{c,t} < 0$, pulling down the dirty good wedge. Intuitively, shifting expenditure from the clean good to the dirty good raises the marginal utility of the clean good, strengthening the incentive to work. This channel counteracts the previous one here, so the net effect of the terms depends on their relative size. Adding back η^s_t to ζ_t, $\eta^s_t < 0$ reinforces the pull, while $\eta^s_t > 0$ weakens or even reverses it.

The fiscal adjustment term to the social cost of carbon also gains two terms. If $\eta_{t+j} > 0$ and having a higher labor income raises the willingness to pay for environmental quality, $\mathcal{I}^S_{y,t+j} > 0$, then the term is positive, pulling up the social cost of carbon. If we set aside the savings multiplier η^s_t in ζ_t such that $\zeta_t > 0$ and if consuming more of the clean good raises the willingness to pay for environmental quality, $\mathcal{I}^S_{c,t+j} > 0$, then the term is once again positive, further pulling up the social cost of carbon. Hence, the planner wants to encourage work and can do so by providing more environmental quality, which high earners and high spenders value more.

Corollary 4.1 (Constrained-optimal carbon price with separable preferences)**.**
Under weak separability or $\gamma_{c,t} = \gamma_{q,t}$,

$$\frac{\tau^q_t - \mathrm{SCC}_t}{1 + \tau^q_t} = -\,\zeta_t\, \mathcal{I}^q_{c,t}$$

When the dirty good is normal, the carbon price is below (above) it where $\zeta_t > 0$ ($\zeta_t < 0$).

Proof. Proposition 4.1 at $\gamma_{c,t} = \gamma_{q,t}$. □

If labor or savings wedges are fixed, then weak separability no longer ensures that the carbon price is set at the social cost of carbon. When consumption moves the willingness to pay for the dirty good, the carbon price can be used to offset the wedge constraints.

4.1.1 Savings-wedge constraint over the life cycle

Whereas the sign of the labor-wedge multiplier is determined by whether labor is underdistorted or overdistorted at a given date and history, acting as a local "push", that of the savings-wedge multiplier pulls in opposite directions at different dates, acting as a "pivot" across the life cycle.

Lemma 4.1 (Sign of the savings multiplier)**.**
Along any history on which η_t^s is nonzero at some date, it must take both positive and negative values over the life cycle. The result holds for any labor schedule and skill process.

Proof. See Appendix B.2. □

Intuitively, consider the case where savings is always overdistorted. If we were talking about the labor wedge, this would only affect consumption-labor choices within a single date. However, the savings wedge here frontloads consumption. Because savings is made to offer a lower return, the household consumes more today and less tomorrow. Therefore, the planner would like to reduce consumption today and increase it tomorrow. Without the means to directly affect the savings wedge, the planner's second option is to affect the consumption mix, if consumption of one good affects demand for the other good. That is, if the dirty good encourages consumption of the clean good, then the planner counteracts the frontloading by distorting the dirty good more today and less tomorrow, shifting clean good consumption from today to tomorrow. This lemma shows that both signs must appear, but does not specify their order or timing.

4.2 Linear dirty good wedge

Aside from income wedge constraints, the planner may be unable to set allocations that imply different dirty good wedges across households, equivalent to observed linear carbon taxes. Suppose first that income wedges are now free, but the allocation must be such that the dirty good wedge is common across all households at a given date. Let $s_{qq,t} < 0$ denote the compensated response of dirty good demand to its price holding labor and pollution fixed.

$$\varpi_t \equiv \frac{\bar{\tau}_t^q - \mathrm{SCC}_t}{1 + \bar{\tau}_t^q} \qquad\qquad \varpi_t^* \equiv (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t}$$

Proposition 4.2 (Optimal linear dirty good wedge)**.**
With income wedges free, the linear dirty good wedge satisfies

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}|\ \varpi_t^*]}{\mathbb{E}_0[|s_{qq,t}|]}$$

Proof. See Appendices C.2 and C.4. □

The optimal dirty good wedge averages the desired corrections to the social cost of carbon, with greater weight on households whose dirty-good demand responds more to a compensated price change. When the desired correction exceeds the common correction, raising the price contributes to reducing the planner's resource cost. When it falls short of this correction, this pulls the price down. The optimum balances this across households. This is similar to the multiple-agent Ramsey rule of Diamond (1975). As shown in Appendix C.4, this restriction can also change the social cost of carbon. A similar rule applies when income wedges are fixed.

Corollary 4.2 (Optimal linear dirty good wedge with fixed income wedges)**.**
Define

$$\varpi_t^{**} \equiv \varpi_t^* + (\gamma_{c,t} - \gamma_{q,t})\,\frac{\eta_t}{y_t} - \zeta_t\,\mathcal{I}_{c,t}^q$$

With labor and savings wedges fixed, the linear dirty good wedge satisfies

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}|\ \varpi_t^{**}]}{\mathbb{E}_0[|s_{qq,t}|]}$$

Proof. See Appendices C.3 and C.4. □

Under weak separability, $\varpi_t^{**} = -\zeta_t \mathcal{I}_{c,t}^q$, so the common wedge equals the social cost of carbon when $\mathbb{E}_0[|s_{qq,t}|\zeta_t \mathcal{I}_{c,t}^q] = 0$. A wedge held constant over the lifecycle must balance marginal resource costs across households as well as dates.

In the next section, I use Proposition 4.2 and Corollary 4.2 to quantify the welfare effects of constraining the dirty good wedge to be linear across households in a single date (age-dependent carbon tax) and also across ages (linear carbon tax), using a model calibrated to US data.

5 Quantitative exercise

The theory suggests that a carbon tax should be nonlinear, but what are the likely welfare gains of implementing more flexible carbon pricing instruments? When income taxes are fixed, is using a nonlinear or even age-dependent carbon tax a good substitute for income

tax reform when the latter is difficult to implement? To answer these questions, I solve the model calibrated to US data, with an energy composite good standing in for the dirty good and other types of consumption standing in for the clean good.

I follow the quantitative setup and resolution method of Farhi and Werning (2013). I extend their logarithmic consumption preferences to two goods, where the two goods are a CES aggregate of both goods with a weight on the energy good that depends on labor supply.

$$U(c,q,l) = \log\left(\left[(1-b_q)c^{\frac{\varsigma-1}{\varsigma}} + b_q e^{-\varkappa l^\alpha} q^{\frac{\varsigma-1}{\varsigma}}\right]^{\frac{\varsigma}{\varsigma-1}}\right) - \frac{l^\alpha}{\alpha}$$

b_q is the weight of energy in the utility function, calibrated to match the observed share of energy in total consumption, about 7% in the 2023 Consumer Expenditure Survey (U.S. Bureau of Labor Statistics 2024). ς is the elasticity of substitution between the two goods, set to 0.4 so that the implied own-price elasticity is about -0.4. West and Williams (2007) report that gasoline is a relative leisure complement (labor substitute), so I set $\varkappa$ to be positive, at 0.25. Households work from age 25 to 64 and are retired from 65 to 84. Since retirees do not work, there is no more screening problem after age 64.

Since a single cohort in a single country cannot accurately capture the social cost of carbon, I fix climate damages at the social cost of carbon, which I set to be consistent with \$190/tCO$_2$ (U.S. Environmental Protection Agency 2023). This means that any deviation from the social cost of carbon is solely due to redistribution and not due to the evolution of climate damages over time. Because the social cost of carbon in the dirty good wedge is common across all households, it affects the level but not the shape of the optimal carbon tax. Hence, this simplification does not affect the qualitative distributional implications.

I compare carbon pricing when income wedges are optimized and when they are fixed. I calibrate the labor wedge using the formula developed in Heathcote et al. (2017) and Heathcote and Tsujiyama (2021), with level parameter $\lambda^T = 0.840$ and progressivity $p = 0.181$. Since there is no HSV-style calibration for the capital income tax, I set a constant savings wedge of 0.9%, equivalent to the average effective tax rate of 18% reported by the CBO[2].

[2]Congressional Budget Office 2014

Table 1: Calibration

Parameter or variable	Value	Calibration basis
Discount factor, β	0.95	$\beta R = 1$, from Farhi and Werning (2013)
Labor-disutility curvature, α	3	Farhi and Werning (2013)
Log-skill innovation variance	0.0095	Farhi and Werning (2013)
Energy expenditure share	7%	U.S. Bureau of Labor Statistics (2024)
Substitution elasticity, ς	0.4	Labandeira et al. (2017)
Leisure complementarity, $\varkappa$	0.25	Sign motivated by West and Williams (2007)
Labor progressivity, p	0.181	Heathcote et al. (2017)
Labor schedule level, λ^T	0.840	Heathcote and Tsujiyama (2021)
Savings wedge, $\bar{\tau}^s$	0.009	Congressional Budget Office (2014)

Under my assumptions, as shown in Figure 1, the planner's preferred labor wedge is much more progressive than the US-calibrated schedule and the planner's preferred savings wedge is decreasing in age. Since it is plausible that the observed wedges can be rationalized with a different calibration on preferences and the skill shock process, I am not making a statement about the optimality of the US tax system. Therefore, I also consider a second exercise where income wedges are fixed at schedules that would have been optimal in a world without externalities. This simulates the case of a government that had already optimized its tax system, but cannot readjust it after introducing carbon taxes. I then compare the welfare gains from using more flexible carbon tax instruments against those of an income tax reform, measured as departures from US-calibrated income wedges towards the planner's preferred wedges.

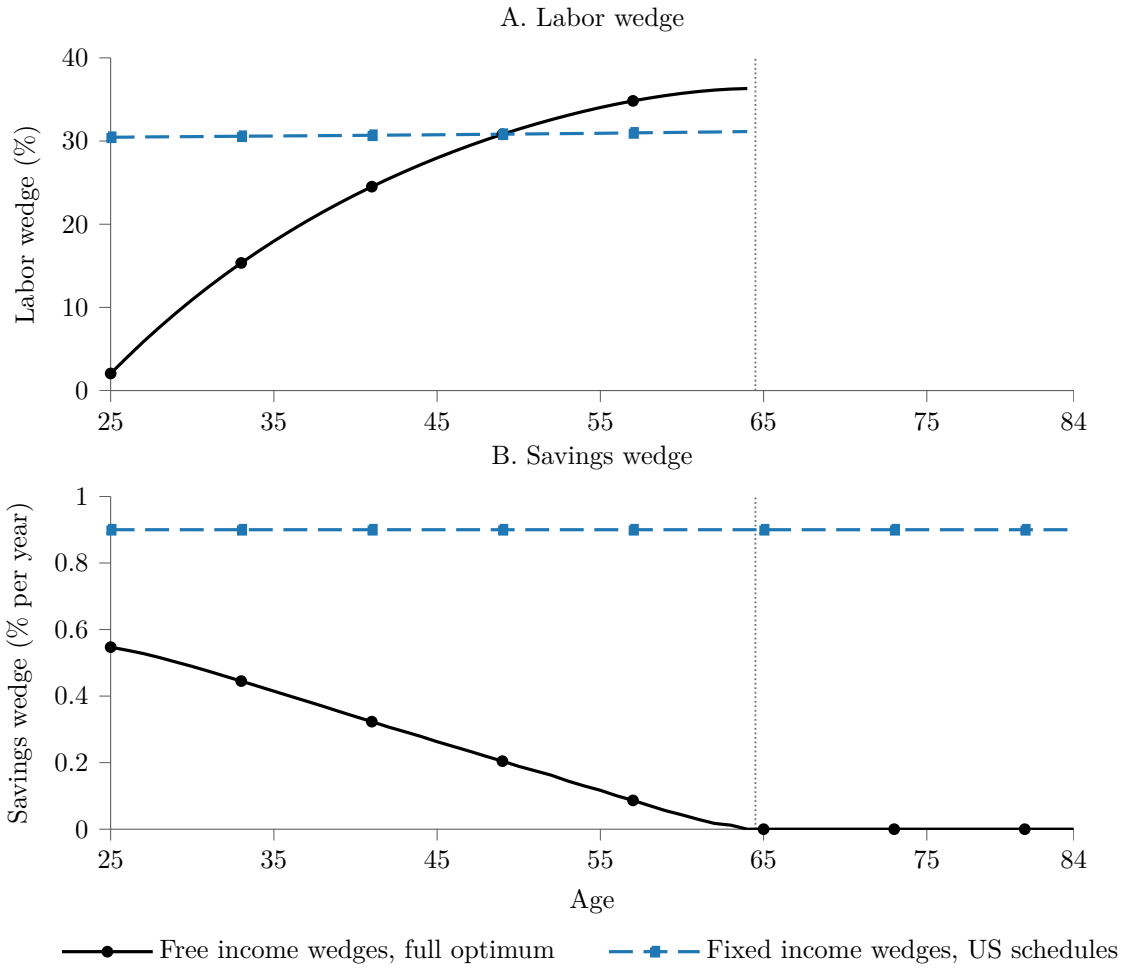


Figure 1: Average labor and savings wedges by age, with income wedges free or fixed.

5.1 Progressivity of fully nonlinear carbon taxes

What does an age-dependent or fully nonlinear carbon tax look like and how progressive is it? Figure 2 plots the average optimal dirty good wedge over the lifecycle, comparing age-dependent and fully nonlinear carbon wedges in a world with free labor and savings wedges and in a world with fixed labor and savings wedges.

In a world with free labor and savings wedges, the average optimal fully nonlinear carbon tax starts slightly above the social cost of carbon of \$190/tCO_2 and rises steadily to about \$245/tCO_2 by the end of the working life. Afterwards, it drops to the social cost of carbon, since households enter retirement and there is no more labor skill screening problem. The best age-dependent carbon tax tracks the average optimal fully nonlinear carbon tax almost perfectly and is therefore not plotted in the figure. Hence, under an optimal allocation, incentive-provision calls for a carbon tax that is higher than the social cost of carbon during the working life.

In a world with fixed labor and savings wedges, the average fully nonlinear carbon tax starts much higher than the social cost of carbon, at a bit over \$250/tCO$_2$, and declines steadily until crossing the social cost of carbon at 34 and then becoming a subsidy from age 62. This pattern comes from the fixed savings wedge. Since the savings wedge is too high at every age, households consume too much early in life, and a carbon tax that falls with age makes later consumption relatively cheaper, which works like a subsidy to saving. The age-dependent carbon tax follows a similar pattern.

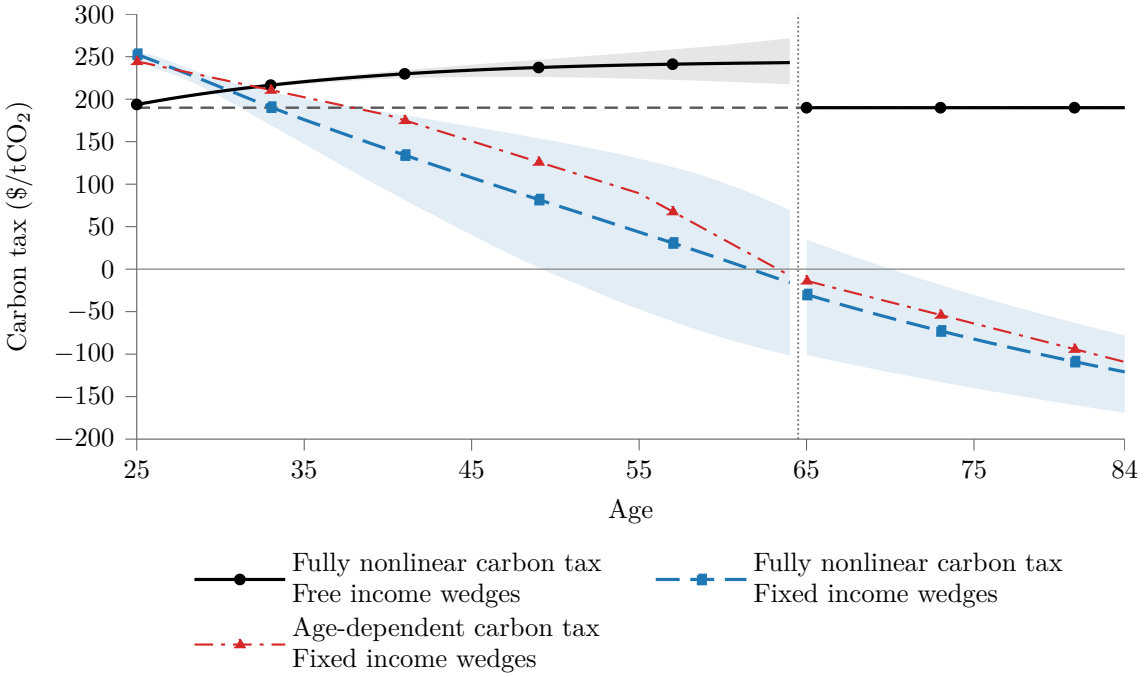


Figure 2: Average carbon tax by age, with cross-household 10th-90th percentile bands. Gray dashes mark the social cost of carbon (\$190/tCO$_2$), dots mark retirement at 65.

Figure 3 shows that the fully nonlinear carbon tax is progressive with respect to current-skill quintile, increasingly so with age. At age 64, in a world with free labor and savings wedges, the average optimal fully nonlinear carbon tax rises from about \$218/tCO$_2$ for the lowest-skill quintile to about \$271/tCO$_2$ for the highest-skill quintile. In a world with fixed labor and savings wedges, the average optimal fully nonlinear carbon tax rises from about −\$99/tCO$_2$ for the lowest-skill quintile to about \$70/tCO$_2$ for the highest-skill quintile. Hence, when income taxes are fixed, the carbon tax is about three times more progressive. While everyone pays a positive tax higher than the social cost of carbon in the first world, the first three quintiles receive a carbon subsidy rather than pay a tax in the second world.

Progressivity builds up over the working life. At age 35, the carbon tax barely varies across quintiles when income wedges are free (about \$218 to \$222/tCO$_2$), but already rises from about \$151 to \$196/tCO$_2$ when they are fixed, and the lowest quintile starts

receiving a subsidy on average at age 50. In short, the optimal carbon tax changes across households and ages, and may even be negative for some.

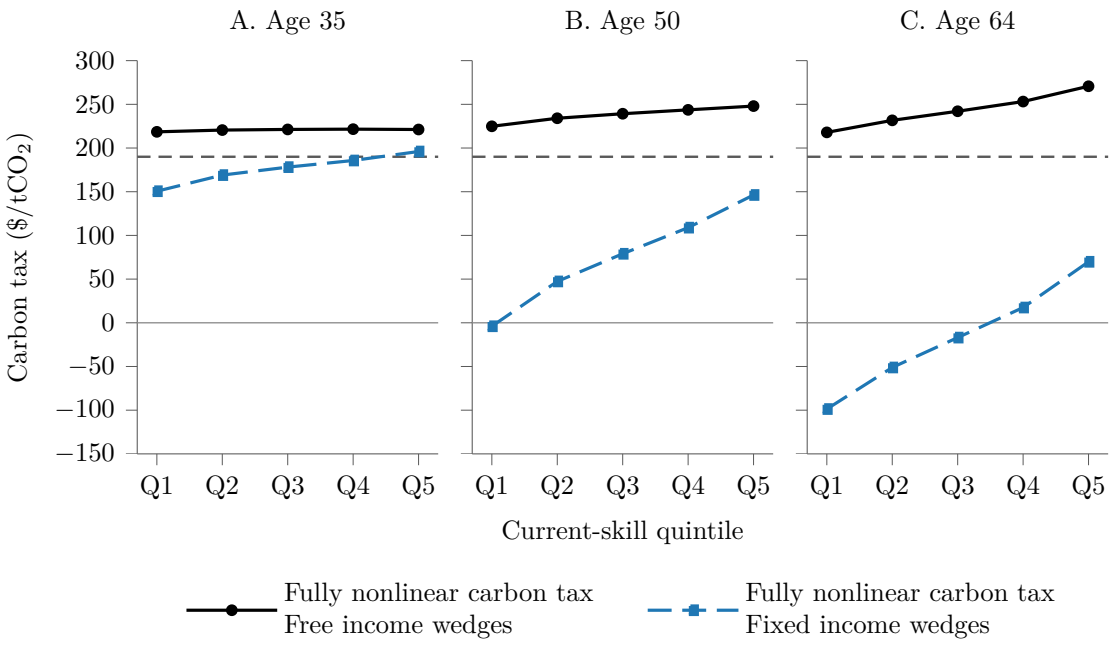


Figure 3: Average fully nonlinear carbon tax by current-skill quintile at ages 35, 50 and 64. Wedges are converted to dollars. Gray dashes mark the social cost of carbon, \$190/tCO$_2$.

These results suggest a significant redistributive role to carbon taxation, but this might be due to the US-calibrated income wedges being far from what the planner would like. Is there still a significant role for redistribution when the carbon tax is introduced to a world with optimized income taxes, but the latter are simply not allowed to readjust afterwards?

5.2 Dirty good wedges alongside previously optimal income wedges

Suppose the government did not use to worry about climate change and optimized its income tax system accordingly. In other words, it did its best given its information. Aftwards, it acknowledges climate change and decides to introduce a carbon tax. However, it cannot reform the entire tax system, meaning labor and savings wedges must remain fixed. Does this lack of coordination call for a change in the way the carbon tax is set?

Table 2: Welfare gains with fixed income wedges

Carbon tax instrument	With climate damages	Without climate damages
Age-dependent carbon tax	0.00037	0.00031
Fully nonlinear carbon tax	0.00056	0.00047

Notes: Consumption-equivalent welfare gains, in % of lifetime consumption, relative to the best linear carbon wedge in each scenario. Labor and savings wedges are fixed at schedules optimized in the economy without climate damages.

Table 2 compares the gains from more flexible carbon instruments when labor and savings wedges are fixed at levels that would have been optimal in a world without externalities.

The best linear rate becomes about \$206/tCO$_2$ when income wedges can adjust, adding only 0.000059% of consumption. At the social cost of carbon, the gain from income wedges reoptimization is below 0.000001%. With income wedges frozen, age-dependent carbon wedges range from about \$192 to \$219/tCO$_2$ (Figure 4). The age-dependent wedge rises during working life and falls close to the social cost of carbon at retirement. Average fully nonlinear carbon wedges follow a similar path. The linear wedge, about \$205/tCO$_2$, captures 99.8% of the gain from introducing the optimal fully nonlinear carbon wedge. Age differentiation adds 0.00037% relative to the linear wedge, about two thirds of the 0.00056% gain from the fully nonlinear carbon wedge.

Having a non-zero dirty good wedge between the two goods is already useful without environmental damages. Without externalities, age differentiation adds 0.000309% relative to the best linear rate. A fully nonlinear wedge adds 0.000470% in welfare.

Hence, designing a carbon tax does not need to be paired with a redesign of the income tax system when it is already optimal. However, if it is not, is allowing for a more flexible carbon tax an adequate substitute for income tax reform?

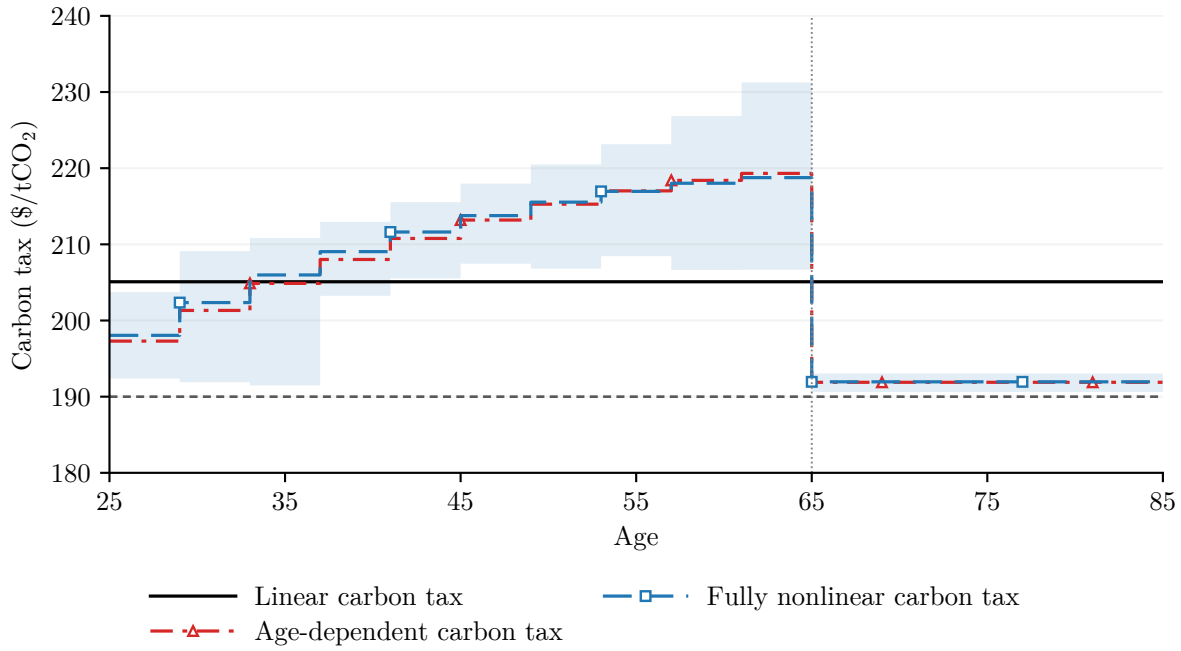


Figure 4: Average optimal carbon taxes with labor and savings wedges fixed at levels that would have been optimal in a world without externalities. Shading shows the 10th-90th percentiles of the fully nonlinear carbon tax. Gray dashes mark the social cost of carbon (\$190/tCO$_2$), dots mark retirement at 65.

5.3 Income tax reform or carbon tax reform?

I come back to the US-calibrated income wedges, which the planner disagrees with. What is the welfare gain from allowing the dirty good wedge to be age-dependent, or even fully nonlinear? How does this compare with the welfare gain from going from the US-calibrated income wedges to the planner's preferred wedges?

Table 3 reports the welfare gains from each reform, compared against a no-tax benchmark in the US-calibrated allocation. In an optimized allocation, adding a dirty good wedge increases welfare by an appreciable 21.5% over the no-carbon benchmark. However, making the dirty good wedge age-dependent or fully nonlinear has little value, as the linear carbon wedge case already captures 99% of the gain from the fully nonlinear carbon wedge. This is because distorting consumption choices has little role in redistribution when income wedges are optimal.

On the other hand, in a US-calibrated allocation, flexibility in the dirty good wedge is more important. Allowing the dirty good wedge to be fully nonlinear increases welfare by 57.1% over the linear carbon wedge, of which the age-dependent carbon wedge captures about 74% of the gain. Hence, even if the fully nonlinear carbon wedge is hard to implement, making it age-dependent gets us most of the way there.

Table 3: Welfare gains from carbon and income tax reform

	Carbon tax instrument			
Income wedges	None	Linear	Age-dependent	Fully nonlinear
US-calibrated	0.0000	0.1913	0.2723	0.3005
Reoptimized	1.4569	1.7706	1.7720	1.7723

Notes: Consumption-equivalent welfare gains, in % of lifetime consumption, relative to US-calibrated income schedules without a carbon tax. Linear carbon wedges are common across households and ages. Age-dependent carbon wedges are common within each age. Fully nonlinear carbon wedges can also differ across households. Labor and savings wedges are fixed in the US-calibrated row and jointly reoptimized in the second row.

However, the gains from more flexible dirty good wedges are dwarfed by those of reoptimizing income wedges. Even holding dirty good wedges linear, going from the US-calibrated income wedges to the planner's preferred wedges adds 1.5793% of consumption, over 16 times the gain from making the dirty good wedge fully nonlinear. Given the large differences between the US-calibrated and planner's preferred income wedges, it might be informative to compare more modest reforms. I consider a 1% decrease in the capital income tax, a 1-2% decrease in the labor income tax, increases in the HSV progressivity parameter, capital income tax cuts in retirement since the model calls for no taxation after retirement, and graduated labor income tax cuts for younger workers since this is where the labor wedge is furthest from the planner's preferred labor

wedge. Because age-dependent carbon taxation delivers most of the welfare of a fully nonlinear carbon tax, I compare these gains against it. Table 4 reports the welfare gains of these limited reforms and the size of the reforms that would match the welfare gains of age-dependent carbon taxation.

Table 4: Welfare gains from limited income tax reforms

Instrument	Reform	Welfare gain	Equivalent reform
Capital income tax	$18\% \to 17\%$	0.2843	$18\% \to 17.1\%$
Labor income tax	-1%	0.2589	-1.2%
	-2%	0.3200	
HSV progressivity (p)	$0.181 \to 0.231$	0.2240	$0.181 \to \approx 0.313$
	$0.181 \to 0.310$	0.2706	
	$0.181 \to 0.350$	0.2963	
Retiree capital income tax	$18\% \to 9\%$	0.2662	$18\% \to \approx 8\%$
	$18\% \to 0\%$	0.3285	
Graduated youth relief	From -2% at 25 to -0.5% at 32–34	0.2770	–

Notes: Welfare gains are in % of lifetime consumption, relative to US-calibrated income schedules without a carbon tax. All income reforms retain a linear carbon wedge of 0.40 (about \$154/tCO_2). Equivalent reforms match the 0.2723% gain from age-dependent carbon taxation. Retirement relief leaves working-life savings wedges unchanged. The youth profile cuts labor income rates by 2, 1.5, 1 and 0.5 points at ages 25, 26-28, 29-31 and 32-34.

The first experiment is the equivalent of a broad 1% capital income tax cut, which delivers a welfare gain of 0.2843%, exceeding the gains of age-dependent carbon taxation. In the model, this is the equivalent of reducing the savings wedge by 0.05%. The second experiment is the equivalent of a broad 1% or 2% labor income tax cut, which delivers a welfare gain of 0.2589% or 0.3200%, respectively. A cut of about 1.2% would match the welfare gain of implementing the best age-dependent carbon tax. The third experiment considers different increases in the HSV progressivity parameter, by 0.05, by 0.129, and by 0.169. The effect is smaller, reaching the welfare gains of an age-dependent carbon tax requiring increasing the parameter by about 0.132, or a 73% increase over the old parameter. The fourth experiment cuts retiree capital income taxes in half and then also considers eliminating them, with a roughly 45% cut being sufficient to match the welfare gains of an age-dependent carbon tax. The fifth experiment implements a graduated youth relief, with cuts of 2, 1.5, 1 and 0.5 points at ages 25, 26-28, 29-31 and 32-34. This particular ladder is illustrative, but is meant to take into account that the difference between US-calibrated and planner's preferred labor wedges is largest at younger ages. The resulting welfare gain is about the same as that of the age-dependent carbon tax.

These examples suggest that small broad-based labor and capital tax cuts can deliver enough welfare to match the gains of an age-dependent carbon tax. If the tax system is particularly suboptimal for some ages, a targeted reform can work too. However, making the labor income tax more progressive requires an implausibly large increase in the progressivity parameter.

In short, even in a world where the income tax system implies wedges that are far from where the planner would want them to be, the welfare gain from implementing an age-dependent carbon tax is small relative to limited income tax reforms. Moreover, non-targeted income tax cuts can still be consistent with the objectives of a redistributive planner when income taxes are overall higher than where they need to be. In the US-calibrated example, labor wedges are highest for younger workers and savings wedges are always too high. The difference between observed and preferred wedges depends on the country and the problem's assumptions, but the takeaway remains that carbon pricing flexibility is a poor substitute for income tax reform.

6 Conclusion

In this paper, I explore the optimal design of carbon taxes when the government's goal is to correct for a climate externality while also redistributing income in a world where households face labor income shocks. In a dynamic Mirrlees model with two goods, I analytically show that whether households should face different carbon taxes depends on how consumption choices interact with labor supply decisions. When the dirty good is a relative labor complement, the optimal carbon tax is lower than the social cost of carbon. When labor and savings wedges are fixed at potentially suboptimal levels, the optimal carbon tax may have further correction terms offsetting these constraints.

The quantitative exercise shows the shape and progressivity of an age-dependent or fully nonlinear carbon wedge, shows that the dirty good wedge does not need to be significantly adjusted even when income wedges were set in a world where the planner did not account for climate externalities and could not be reoptimized, and that the welfare gains from making the dirty good wedge age-dependent or fully nonlinear are small even relative to modest income tax reforms. Although the theory calls for nonlinear history-dependent carbon taxes, the numerical results suggest that the redistributive role of carbon taxation is limited.

Hence, at least within the confines of a Mirrlees type analysis, there is little practical benefit to deviating from the standard Pigouvian rule of setting a linear carbon tax equal to the social cost of carbon, with redistributive goals best achieved with limited income tax reform. If income taxes cannot change at all, an age-dependent carbon tax can capture

most of the gains of a fully nonlinear carbon tax. In practice, this could for example be implemented by a linear carbon tax with age-dependent income-agnostic rebates.

This paper has a number of limitations that would be interesting to address in future work. First, it focuses on allocations rather than tax implementations, so it cannot speak to the specifics of how an age-dependent or even fully nonlinear carbon tax could be best put in place. Moreover, because carbon tax revenue is implicitly optimally recycled, the model cannot speak as to the best way to use it. Second, the model features a single cohort, so a natural extension is considering an overlapping generations model to study intergenerational redistribution. This would also allow for the implementation of the framework in a full integrated assessment model, in order to better study the implications of income inequality on emissions.

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A Optimal wedge derivations

A.1 First-order conditions

Because $U(c, q, l, S)$ can be written as $U(c, q, y/\theta, S)$, I write U_θ as $-U_l l/\theta$. To derive the first-order conditions of the planner's problem, I keep the continuation value at the population level. At each preceding history θ^{t-1}, write $f = f_t(\theta|\theta_{t-1})$ and $f_2 = \frac{\partial f_t}{\partial \theta_{t-1}}$. The multipliers $\lambda_{1,t}$ and $\lambda_{2,t}$ depend on the preceding history, while ϕ_t and ψ_t also depend on current skill. I normalize these multipliers relative to the probability measure dF^{t-1} and suppress history and time arguments where there is no ambiguity.

Let ν_t be the multiplier on aggregate emissions and let μ_t be the shadow cost of the current pollution stock, including its effects on future pollution stocks. The Lagrangian is

$$\begin{aligned}\mathcal{L}_t = \int \Bigg\{ \int_0^\infty \Bigg(& [c + (1+\nu_t)q - \theta l] f \\ & + \psi \left[-U_l \frac{l}{\theta} + \beta w_2 - \dot{u} \right] - \lambda_1 u \bar{\alpha}_t f + \lambda_2 u f_2 \\ & + \phi[u - U - \beta w] \Bigg) d\theta + \lambda_1 \hat{w}_t - \lambda_2 \hat{w}_{2,t} \Bigg\} dF^{t-1} \\ & + \frac{1}{R} V_{t+1}(w_t(\cdot), w_{2,t}(\cdot), \mathcal{H}_{t+1}) \\ & - \nu_t Q_t - \mu_t [S_t - \mathcal{S}(\mathcal{H}_t, Q_t)]\end{aligned}$$

For a variation δu in remaining lifetime utility, integrating the $-\psi\, \delta\dot{u}$ term by parts gives

$$\delta \mathcal{L}_t = \int \left\{ -[\psi\, \delta u]_0^\infty + \int_0^\infty [\dot{\psi} + \phi - \lambda_1 \bar{\alpha}_t f + \lambda_2 f_2] \delta u \, d\theta \right\} dF^{t-1}$$

The first order conditions are:

$$\begin{aligned}\frac{\partial \mathcal{L}_t}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} = \phi U_c \\ \frac{\partial \mathcal{L}_t}{\partial q} &: (1+\nu_t) f - \psi U_{ql} \frac{l}{\theta} = \phi U_q \\ \frac{\partial \mathcal{L}_t}{\partial l} &: -\theta f - \phi U_l = \frac{1}{\theta} \psi \left[U_{ll} l + U_l \right] \\ \frac{\partial \mathcal{L}_t}{\partial u} &: \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi\end{aligned}$$

For $t < T$, consider variations δw_t and $\delta w_{2,t}$ in the continuation promises. Holding $\mathcal{H}_{t+1}$, the envelope theorem applied to next period's problem gives

$$\delta V_{t+1} = \int [\lambda_{1,t+1}\delta w_t - \lambda_{2,t+1}\delta w_{2,t}] dF^t$$

Since $dF^t = f_t \, d\theta \, dF^{t-1}$, the corresponding variation of the Lagrangian is

$$\begin{aligned}\delta \mathcal{L}_t &= \int \int_0^\infty [-\beta \phi_t \delta w_t + \beta \psi_t \delta w_{2,t}] d\theta \, dF^{t-1} + \frac{1}{R}\delta V_{t+1} \\ &= \int \left[\left(\frac{\lambda_{1,t+1}}{R} - \frac{\beta \phi_t}{f_t} \right) \delta w_t + \left(\frac{\beta \psi_t}{f_t} - \frac{\lambda_{2,t+1}}{R} \right) \delta w_{2,t} \right] dF^t\end{aligned}$$

At the optimum, the coefficients of these independent variations are zero when $f_t > 0$, so

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi_t(\theta^t)}{f_t(\theta_t|\theta_{t-1})} \qquad \lambda_{2,t+1}(\theta^t) = \beta R \frac{\psi_t(\theta^t)}{f_t(\theta_t|\theta_{t-1})} \tag{13}$$

Pollution affects all households. If we differentiate the Lagrangian with respect to S_t,

$$\mu_t = -\int \left[\frac{\psi_t}{f_t} U_{Sl,t} \frac{l_t}{\theta_t} + \frac{\phi_t}{f_t} U_{S,t} \right] dF^t + \frac{1}{R} \frac{\partial V_{t+1}}{\partial S_t} \tag{14}$$

Let $S_q \equiv \frac{\partial \mathcal{S}}{\partial Q_t}$ be the marginal effect of current emissions on the current pollution stock. Holding the promise functions and S_t fixed, differentiating with respect to Q_t gives

$$\frac{\partial \mathcal{L}_t}{\partial Q_t} = -\nu_t + \mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q_t} = 0 \qquad \Longrightarrow \qquad \nu_t = \mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q_t} \tag{15}$$

The three wedges can be derived from here.

A.2 Labor wedge

Dividing the first-order condition for c of Appendix A.1 by U_c and using $\gamma_c \equiv U_{cl} l / U_c$,

$$\phi = \frac{f}{U_c} - \frac{\psi}{\theta} \gamma_c \tag{16}$$

From the first-order condition for l and since $U_{ll} l + U_l = U_l(1 + \epsilon)$ with $\epsilon \equiv U_{ll} l / U_l$,

$$-\theta f - \phi U_l = \frac{\psi}{\theta} U_l (1+\epsilon) \implies \phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1+\epsilon)$$

Equating the two expressions for ϕ and collecting the terms in ψ,

$$\frac{f}{U_c} + \frac{\theta f}{U_l} = -\frac{\psi}{\theta}(1+\epsilon-\gamma_c)$$

The labor wedge is $1-\tau^y = -U_l/(\theta U_c)$, so $\theta U_c/U_l = -1/(1-\tau^y)$, making the left-hand side $\frac{f}{U_c}\left(1 - \frac{1}{1-\tau^y}\right) = -\frac{f}{U_c}\frac{\tau^y}{1-\tau^y}$. Substituting and rearranging,

$$\frac{\tau^y}{1-\tau^y} = (1+\epsilon-\gamma_c)\frac{\psi U_c}{\theta f} \tag{17}$$

Next, substitute ϕ from (16) into $\dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi$,

$$\dot{\psi} - \frac{\gamma_c}{\theta}\psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}$$

This is a linear first-order ordinary differential equation. For this derivation, I assume at each date and history that $\lim_{\theta \downarrow 0} \psi(\theta) = \lim_{\theta \to \infty} \psi(\theta) = 0$ and that $\gamma_c(\theta)/\theta$ is absolutely integrable near zero. I also assume that each term in the weighted integrals below is absolutely integrable and that the denominator in (19) is positive.

I use the integrating factor $\exp\left[-\int_0^\theta \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]$. Integrating from θ to $b > \theta$ and dividing by the factor at θ leaves the upper boundary term $\exp\left[-\int_\theta^b \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\psi(b)$. I impose the weighted boundary condition

$$\lim_{b\to\infty} \exp\left[-\int_0^b \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\psi(b) = 0$$

For any fixed $\theta > 0$, this makes the upper boundary term disappear. Taking $b \to \infty$ gives

$$\psi(\theta) = \int_\theta^\infty \exp\left[-\int_\theta^x \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x)\right) dx \tag{18}$$

Absolute integrability near zero makes the integrating factor converge to one as θ approached 0. Absolute convergence of the weighted integrals then allows taking this limit in (18). Using $\lim_{\theta \to 0} \psi(\theta) = 0$ and rearranging gives

$$\lambda_{1,t} = \frac{\int_0^\infty \exp\left[-\int_0^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{1}{U_{c,t}(x)}f_t(x) + \lambda_{2,t}f_{2,t}(x)\right)dx}{\int_0^\infty \exp\left[-\int_0^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\bar{\alpha}_t(x)f_t(x)dx} \tag{19}$$

Substituting (18) into (17),

$$\frac{\tau^y}{1-\tau^y} = (1+\epsilon-\gamma_c)\frac{U_c(\theta)}{\theta f(\theta)}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right]\left(\frac{f(x)}{U_c(x)} - \lambda_1\bar{\alpha}_t(x)f(x) + \lambda_2 f_2(x)\right)dx$$

Next, I rewrite the first part of the integral using

$$\frac{d}{d\theta}\ln U_{c,t} = \frac{U_{cc}\dot{c} + U_{cq}\dot{q} + U_{cl}\dot{l}}{U_c} = -\sigma_t\frac{\dot{c}}{c} - \sigma_{cq,t}\frac{\dot{q}}{q} + \gamma_c\frac{\dot{l}}{l}$$

Since $l = y/\theta$ gives $\dot{l}/l = \dot{y}/y - 1/\theta$, integrating from θ to x and multiplying by the integrating factor gives the following:

$$\frac{U_{c,t}(\theta)}{U_{c,t}(x)}\exp\left[-\int_\theta^x \gamma_c(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] = \exp\left[\int_\theta^x\left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_c(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right)d\tilde{x}\right]$$

The first part of the labor wedge is therefore

$$(1+\epsilon_t(\theta)-\gamma_{c,t}(\theta))\frac{1}{\theta f_t(\theta)}\int_\theta^\infty \exp\left[\int_\theta^x\left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right)d\tilde{x}\right]$$
$$\left(1-\lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x)\right)f_t(x)\,dx$$

For the second part of the labor wedge, use (13) and shift the date back one period,

$$\lambda_{2,t}(\theta^{t-1}) = \beta R\frac{\psi_{t-1}}{f_{t-1}}$$

Next, evaluating (17) at $t-1$,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\frac{\theta_{t-1}}{U_{c,t-1}\left(1+\epsilon_{t-1}-\gamma_{c,t-1}\right)}$$

The $\lambda_2 f_2$ term of the wedge is $\lambda_{2,t}$, a constant in the current type, times the same terms before the term in parentheses.

$$(1+\epsilon_t(\theta)-\gamma_{c,t}(\theta))\,\frac{U_{c,t}(\theta)}{\theta f_t(\theta)}\,\lambda_{2,t}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\,dx$$

Substituting the two and rearranging,

$$\beta R\,\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}\,\frac{1+\epsilon_t(\theta)-\gamma_{c,t}(\theta)}{1+\epsilon_{t-1}-\gamma_{c,t-1}}\,\frac{U_{c,t}(\theta)}{U_{c,t-1}}\,\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)\,dx}{\theta f_t(\theta)}$$

Putting together the two and multiplying and dividing the first term by $1-F_t(\theta)$,

$$\frac{\tau^y_t}{1-\tau^y_t} = A_t(\theta)B_t(\theta)C_t(\theta) + \beta R\frac{\tau^y_{t-1}}{1-\tau^y_{t-1}}D_t(\theta)$$

with

$$A_t(\theta) = 1+\epsilon_t(\theta)-\gamma_{c,t}(\theta)$$
$$B_t(\theta) = \frac{1-F_t(\theta)}{\theta f_t(\theta)}$$
$$C_t(\theta) = \int_\theta^\infty \exp\left[\int_\theta^x \left(\sigma_t(\tilde{x})\frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x})\frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x})\frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})}\right)d\tilde{x}\right](1-\lambda_{1,t}\bar{\alpha}_t(x)U_{c,t}(x))\,\frac{f_t(x)dx}{1-F_t(\theta)}$$
$$D_t(\theta) = \frac{A_t(\theta)}{A_{t-1}}\frac{U_{c,t}(\theta)}{U_{c,t-1}}\frac{\theta_{t-1}\int_\theta^\infty \exp\left[-\int_\theta^x \gamma_{c,t}(\tilde{x})\frac{d\tilde{x}}{\tilde{x}}\right] f_{2,t}(x)dx}{\theta f_t(\theta)}$$

A.3 Savings wedge

The savings wedge follows from the same first-order conditions of Appendix A.1. I define χ_t, a term reducing the cost of delivering utility when consumption complements labor.

$$\chi_t(\theta^t) \equiv \frac{\gamma_{c,t}}{1+\epsilon_t-\gamma_{c,t}}\frac{\tau^y_t}{1-\tau^y_t}$$

Using $\frac{\tau^y_t}{1-\tau^y_t} = \frac{\psi U_c}{\theta f}(1+\epsilon_t-\gamma_{c,t})$, $\phi = f/U_c - \psi\gamma_c/\theta$ from (16) can be rewritten as follows:

$$\frac{\phi}{f} = \frac{1}{U_c}\left(1-\gamma_c\frac{\psi U_c}{\theta f}\right) = \frac{1-\chi_t}{U_{c,t}} \tag{20}$$

$\chi_t > 0$ when the numéraire is complementary with labor and labor is distorted and zero when preferences are separable between the numéraire and labor or when labor is undistorted. The utility-promise condition in (13) gives

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1-\chi_t}{U_{c,t}} \tag{21}$$

Using the endpoint limits in Appendix A.2 and assuming the integrals below converge,

$$0 = \int_0^\infty \dot{\psi}\, d\theta = \lambda_{1,t} \int_0^\infty \bar{\alpha}_t f\, d\theta - \lambda_{2,t} \int_0^\infty f_2\, d\theta - \int_0^\infty \phi\, d\theta$$

For $t \geq 1$, $\bar{\alpha}_t = 1$ and $\int_0^\infty f_2\, d\theta = 0$, so $\lambda_{1,t} = \int_0^\infty \phi\, d\theta$, which by (20) is

$$\lambda_{1,t} = \mathbb{E}_{t-1}\left[\frac{1-\chi_t}{U_{c,t}}\right] \tag{22}$$

For $t < T$, evaluating (22) at $t+1$ and equating with (21) gives:

$$\frac{1-\chi_t}{U_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1-\chi_{t+1}}{U_{c,t+1}}\right]$$

Where $1-\chi_t > 0$ at the current history and $1-\chi_{t+1} > 0$ at every next-period history, defining the incentive-adjusted marginal utility $\widehat{U}_{c,t} \equiv U_{c,t}/(1-\chi_t)$,

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R}\,\mathbb{E}_t\left[\frac{1}{\widehat{U}_{c,t+1}}\right]$$

A.4 dirty good wedge

Starting from the first-order conditions of Appendix A.1,

$$\phi = \frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}$$

Because c and q share the same ϕ,

$$\phi = \frac{f}{U_q} - \frac{\psi U_{ql}\frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q}$$

Hence,

$$\frac{f}{U_q} - \frac{\psi U_{ql}\frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q} = \frac{f}{U_c} - \frac{\psi U_{cl}\frac{l}{\theta}}{U_c}$$

Multiplying by U_q and U_c,

$$U_c\left(f-\psi U_{ql}\frac{l}{\theta}+\nu_t f\right)=U_q\left(f-\psi U_{cl}\frac{l}{\theta}\right)$$

Using $\gamma_c \equiv \frac{U_{cl}\,l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql}\,l}{U_q}$ or, equivalently, $\gamma_c U_c = U_{cl}\,l$ and $\gamma_q U_q = U_{ql}\,l$,

$$U_c\left(f-\frac{\psi}{\theta}\gamma_q U_q+\nu_t f\right)=U_q\left(f-\frac{\psi}{\theta}\gamma_c U_c\right)$$

Then, putting the γ terms (as well as the externality term) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta}U_c U_q(\gamma_q-\gamma_c)-\nu_t U_c f$$

Finally, dividing by $U_q f$,

$$\frac{U_c}{U_q}-1=\frac{\psi U_c}{\theta f}(\gamma_q-\gamma_c)-\nu_t\frac{U_c}{U_q}$$

Note that, by definition, $1+\tau^q = U_q/U_c$, so $\frac{U_c}{U_q}-1=-\frac{\tau^q}{1+\tau^q}$. Multiplying through by -1,

$$\frac{\tau^q}{1+\tau^q}=\frac{\psi U_c}{\theta f}(\gamma_c-\gamma_q)+\nu_t\frac{U_c}{U_q}$$

Since $\frac{U_c}{U_q}=\frac{1}{1+\tau^q}$,

$$\frac{\tau^q}{1+\tau^q}=\frac{\psi U_c}{\theta f}(\gamma_c-\gamma_q)+\frac{\nu_t}{1+\tau^q}$$

Rearranging,

$$\frac{\tau^q-\nu_t}{1+\tau^q}=\frac{\psi U_c}{\theta f}(\gamma_c-\gamma_q)$$

Note that, from the labor wedge derivation,

$$\frac{\psi U_c}{\theta f}=\frac{1}{1+\epsilon_t-\gamma_{c,t}}\frac{\tau_t^y}{1-\tau_t^y}$$

Substituting,

$$\frac{\tau_t^q-\nu_t}{1+\tau_t^q}=\frac{\gamma_{c,t}-\gamma_{q,t}}{1+\epsilon_t-\gamma_{c,t}}\frac{\tau_t^y}{1-\tau_t^y}$$

A.5 The social cost of carbon

Substituting $\phi = f/U_c - \psi U_{cl} l/(\theta U_c)$ from the clean good condition into (14) and moving the continuation term to the left-hand side gives

$$\mu_t - \frac{1}{R}\frac{\partial V_{t+1}}{\partial S_t} = \int \left(-\frac{\psi}{\theta f} U_{Sl} l - \left[1 - \frac{\psi}{\theta f} U_{cl} l \right] \frac{U_S}{U_c} \right) dF^t$$

Rewrite $\frac{\psi}{\theta f}$ using (17),

$$\frac{\psi}{\theta f} = \left(\frac{\tau^y}{1-\tau^y} \right) \frac{1}{U_c} \frac{1}{1+\epsilon-\gamma_c}$$

Hence, also defining $\xi \equiv \frac{U_{Sl} l}{U_S}$,

$$\begin{aligned}
\mu_t - \frac{1}{R}\frac{\partial V_{t+1}}{\partial S_t} &= \int \left(-\left(\frac{\tau^y}{1-\tau^y} \right) \frac{1}{1+\epsilon-\gamma_c} \frac{U_{Sl} l}{U_c} - \left[1 - \left(\frac{\tau^y}{1-\tau^y} \right) \frac{1}{1+\epsilon-\gamma_c} \frac{U_{cl} l}{U_c} \right] \frac{U_S}{U_c} \right) dF^t \\
&= \int \left(-\left(\frac{\tau^y}{1-\tau^y} \right) \frac{1}{1+\epsilon-\gamma_c} \frac{U_{Sl} l}{U_c} \frac{U_S}{U_S} - \left[1 - \left(\frac{\tau^y}{1-\tau^y} \right) \frac{\gamma_c}{1+\epsilon-\gamma_c} \right] \frac{U_S}{U_c} \right) dF^t \\
&= \int \left(-\left(\frac{\tau^y}{1-\tau^y} \right) \frac{\xi}{1+\epsilon-\gamma_c} \frac{U_S}{U_c} - \left[1 - \left(\frac{\tau^y}{1-\tau^y} \right) \frac{\gamma_c}{1+\epsilon-\gamma_c} \right] \frac{U_S}{U_c} \right) dF^t \\
&= -\int \left[1 + \left(\frac{\tau^y}{1-\tau^y} \right) \frac{\xi-\gamma_c}{1+\epsilon-\gamma_c} \right] \frac{U_S}{U_c} \, dF^t
\end{aligned}$$

Each integral runs over the date-t population of complete skill histories. The household quantities and multipliers depend on these histories while μ_t is common to the population.

Both the current pollution stock S_t and current emissions Q_t can affect future pollution. Write $\mathcal{S}_j = \mathcal{S}(\mathcal{H}_j, Q_j)$ for pollution stock accumulation at date j. By the envelope theorem, the effect of past pollution stocks or emissions on the continuation value is the discounted sum of its direct effects on later accumulation constraints. For $X_t \in \{S_t, Q_t\}$,

$$\frac{\partial V_{t+1}}{\partial X_t} = \sum_{j=t+1}^{T} \frac{1}{R^{j-t-1}} \mu_j \frac{\partial \mathcal{S}_j}{\partial X_t}$$

Each derivative of $\mathcal{S}_j$ holds its other arguments fixed. Substituting into (15) gives

$$\nu_t = \sum_{j=t}^{T} \frac{1}{R^{j-t}} \mu_j \frac{\partial \mathcal{S}_j}{\partial Q_t} \tag{23}$$

At the terminal date, $V_{T+1} = 0$, so μ_T equals the current-date integral. At earlier dates, μ_t also includes the future costs caused by the persistence of the current pollution stock. By the chain rule,

$$\frac{\partial S_j}{\partial Q_t} = \frac{\partial \mathcal{S}_j}{\partial Q_t} + \sum_{k=t}^{j-1} \frac{\partial \mathcal{S}_j}{\partial S_k} \frac{\partial S_k}{\partial Q_t} \qquad j \geq t \tag{24}$$

The sum is empty at $j = t$, so $\partial S_t / \partial Q_t = S_q$. Subtracting the continuation term from each pollution stock shadow cost, multiplying by $\frac{1}{R^{j-t}} \partial S_j / \partial Q_t$, and summing over j,

$$\begin{aligned}
\sum_{j=t}^{T} \frac{1}{R^{j-t}} \left(\mu_j - \frac{1}{R} \frac{\partial V_{j+1}}{\partial S_j} \right) \frac{\partial S_j}{\partial Q_t} &= \sum_{j=t}^{T} \frac{\mu_j}{R^{j-t}} \left[\frac{\partial S_j}{\partial Q_t} - \sum_{k=t}^{j-1} \frac{\partial \mathcal{S}_j}{\partial S_k} \frac{\partial S_k}{\partial Q_t} \right] \\
&= \sum_{j=t}^{T} \frac{\mu_j}{R^{j-t}} \frac{\partial \mathcal{S}_j}{\partial Q_t} \\
&= \nu_t
\end{aligned}$$

The first equality uses the continuation envelope condition and reorders the finite sums. The second uses (24) and the last follows from (23). Substituting the current-date expression above into the left-hand side gives

$$\mathrm{SCC}_t \equiv \nu_t = -\sum_{j=0}^{T-t} \frac{1}{R^j} \int \left(1 + \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \frac{\tau^y_{t+j}}{1 - \tau^y_{t+j}} \right) \frac{U_{S,t+j}}{U_{c,t+j}} \frac{\partial S_{t+j}}{\partial Q_t} \, dF^{t+j}$$

B Derivations with fixed income wedges

B.1 Constrained-optimal dirty good wedge

The labor-wedge constraint involves only the current allocation and pollution stock, so it adds a pointwise multiplier but no new state. For savings, define the following for $t = 0, \ldots, T-1$:

$$M(\theta^t) \equiv \frac{U_{c,t}(\theta^t)}{(1 - \bar{\tau}_t^s(\theta_t))\beta R}$$

The savings-wedge constraint requires $\mathbb{E}_t[U_{c,t+1}] = M(\theta^t)$. Last period's problem brings $\hat{M}$, which enters as a new state.

$$\hat{M} = \int U_c \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) f_t(\theta \mid \theta_-) \, d\theta$$

Because there is no previous period at birth and no future period at death, $\vartheta_0 = \vartheta_{T+1} = 0$. The constrained problem adds the two wedge constraints to the problem in Appendix A.1.

$$\begin{aligned}
\mathscr{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, M(c, q, l, S), \theta, \mathbf{S}) \right] f \\
& + \psi \left[-U_l \frac{l}{\theta} + \beta w_2 \right] - \lambda_1 u \bar{\alpha}_t f + \lambda_2 u f_2 + \phi[u - U - \beta w] - \mu[S - \mathcal{S}(\ldots, Q)] \\
& + \eta \left[-\frac{U_l}{\theta U_c} - (1 - \bar{\tau}_t^y(\theta)) \right] f + \vartheta_t (U_c - \hat{M}) f
\end{aligned}$$

For $z \in \{c, q, l, S\}$, the new constraints and the modified continuation value contribute

$$\left[\vartheta_t - \frac{\vartheta_{t+1}}{(1 - \bar{\tau}_t^s)\beta R^2} \right] U_{cz} f = \frac{\eta_t^s}{U_c} U_{cz} f$$

From here,

$$
\begin{aligned}
\frac{\mathscr{H}}{\partial c}: &\quad f - \psi U_{cl}\frac{l}{\theta} + \eta f \Gamma_c + \frac{\eta_t^s}{U_c} U_{cc} f = \phi U_c \\
\frac{\mathscr{H}}{\partial q}: &\quad (1+\nu_t) f - \psi U_{ql}\frac{l}{\theta} + \eta f \Gamma_q + \frac{\eta_t^s}{U_c} U_{cq} f = \phi U_q \\
\frac{\mathscr{H}}{\partial l}: &\quad -\theta f - \phi U_l + \eta f \Gamma_l + \frac{\eta_t^s}{U_c} U_{cl} f = \frac{\psi}{\theta}(U_{ll} l + U_l) \\
\frac{\mathscr{H}}{\partial u}: &\quad \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\
\frac{\mathscr{H}}{\partial w}: &\quad \frac{1}{R} V_{t+1,w} f = \phi \beta \\
\frac{\mathscr{H}}{\partial w_2}: &\quad \frac{1}{R} V_{t+1,w_2} f = -\psi \beta \\
\frac{\mathscr{H}}{\partial S}: &\quad \mathbb{E}_0\left[-\frac{\psi}{f} U_{Sl}\frac{l}{\theta} - \frac{\phi}{f} U_S + \eta \Gamma_S + \frac{\eta_t^s}{U_c} U_{cS} \right] = \mu_t - \frac{1}{R}\frac{\partial V_{t+1}}{\partial S_t}
\end{aligned}
$$

Here, $\nu_t = \mu_t S_q + \frac{1}{R}\frac{\partial V_{t+1}}{\partial Q_t}$. The pollution condition averages component contributions over all date-t histories. The continuation derivatives hold the promise functions, M, and other inherited pollution arguments fixed. The effect through M is already included in η_t^s. At the constrained allocation, $\Gamma_z = -U_{lz}/(\theta U_c) - (1-\bar{\tau}_t^y) U_{cz}/U_c$. Hence,

$$
\begin{aligned}
\Gamma_c &= (1-\bar{\tau}_t^y)\frac{\sigma}{c} - \frac{\gamma_c}{y} \\
\Gamma_q &= (1-\bar{\tau}_t^y)\frac{\sigma_{cq}}{q} - \frac{\gamma_q(1+\tau^q)}{y} \\
\Gamma_l &= (1-\bar{\tau}_t^y)\frac{\epsilon - \gamma_c}{l} \\
\Gamma_S &= -\frac{U_S}{U_c}\left[\frac{\xi}{y} + (1-\bar{\tau}_t^y)\frac{\xi_c}{c} \right]
\end{aligned}
$$

Eliminating ϕ between the consumption and dirty-good conditions gives

$$
\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t})\frac{\psi U_{c,t}}{\theta f_t} - \eta_t \Gamma_{c,t} + \eta_t^s \frac{\sigma_t}{c_t} + \frac{\eta_t \Gamma_{q,t} - \eta_t^s \frac{\sigma_{cq,t}}{q_t}}{1+\tau_t^q}
$$

Putting the η_t and η_t^s terms together,

$$
\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t})\frac{\psi U_{c,t}}{\theta f_t} - \eta_t \left[\Gamma_{c,t} - \frac{\Gamma_{q,t}}{1+\tau_t^q} \right] + \eta_t^s \left[\frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t(1+\tau_t^q)} \right]
$$

Using the definitions for Γ_c and Γ_q,

$$\Gamma_{c,t} - \Gamma_{q,t} = (1 - \bar{\tau}_t^y) \left[\frac{\sigma_{c,t}}{c_t} - \frac{\sigma_{cq,t}}{q_t(1+\tau_t^q)} \right] - \frac{\gamma_{c,t} - \gamma_{q,t}}{y_t}$$

Note that the first term in the brackets is the same as the η_t^s term. If we put them together,

$$\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \frac{\gamma_{c,t} - \gamma_{q,t}}{y_t} - ((1 - \hat{\tau}_t^y)\eta_t - \eta_t^s) \left[\frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t(1+\tau_t^q)} \right]$$

Define $\mathcal{I}_{c,t}^q = \frac{\partial}{\partial c_t} \ln\left(\frac{U_{q,t}}{U_{c,t}}\right)$ and $\mathcal{I}_{y,t}^q = \frac{\partial}{\partial y_t} \ln\left(\frac{U_{q,t}}{U_{c,t}}\right)$, representing how the willingness-to-pay for the dirty good responds to changes in clean good consumption and in labor income, respectively. Note that, using the elasticity definitions from the main text,

$$\begin{aligned}
\frac{\partial}{\partial c_t} \ln\left(\frac{U_{q,t}}{U_{c,t}}\right) &= \frac{U_{cq,t}}{U_{q,t}} - \frac{U_{cc,t}}{U_{c,t}} = \frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t} \frac{U_{c,t}}{U_{q,t}} = \frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t(1+\tau_t^q)} \\
\frac{\partial}{\partial y_t} \ln\left(\frac{U_{q,t}}{U_{c,t}}\right) &= \frac{U_{qy,t}}{U_{q,t}} - \frac{U_{cy,t}}{U_{c,t}} = \frac{1}{\theta}\left(\frac{U_{ql,t}}{U_{q,t}} - \frac{U_{cl,t}}{U_{c,t}}\right) = -\frac{\gamma_{c,t} - \gamma_{q,t}}{y_t}
\end{aligned}$$

Defining $\zeta_t = (1 - \hat{\tau}_t^y)\eta_t - \eta_t^s$ and using the above equations,

$$\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} - \eta_t \mathcal{I}_{y,t}^q - \zeta_t \mathcal{I}_{c,t}^q$$

$\frac{\psi U_{c,t}}{\theta f_t}$ is proportional to the labor wedge the planner would normally want to set. Hence, using the labor wedge derivations, we can also write

$$\frac{\tau_t^q - \nu_t}{1+\tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t \mathcal{I}_{y,t}^q - \zeta_t \mathcal{I}_{c,t}^q$$

The only remaining thing to calculate is the current-date component of the stock cost. Using $U_{Sl} l = \xi U_S$ and $U_{cS}/U_c = \xi_c (U_S/U_c)/c$,

$$-\frac{U_S}{U_c} \left[1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta \frac{\xi - \gamma_c}{y} + ((1 - \bar{\tau}_t^y)\eta - \eta_t^s) \frac{\sigma + \xi_c}{c} \right]$$

Define $\mathcal{I}_{c,t}^S = \frac{\partial}{\partial S_t} \ln\left(\frac{U_{q,t}}{U_{c,t}}\right)$ and $\mathcal{I}_{y,t}^S = \frac{\partial}{\partial S_t} \ln\left(\frac{U_{q,t}}{U_{c,t}}\right)$, representing how the willingness-to-pay

for environmental quality responds to changes in clean good consumption and in labor income. From here,

$$\begin{aligned}\mu_t - \frac{1}{R}\frac{\partial V_{t+1}}{\partial S_t} &= -\mathbb{E}_0\left[\left(1 + (\xi - \gamma_c)\frac{\psi U_c}{\theta f} + \eta \mathcal{I}_y^S + \zeta_t \mathcal{I}_c^S\right)\frac{U_S}{U_c}\right] \\ &= -\mathbb{E}_0\left[\left(1 + \frac{\xi - \gamma_c}{1 + \epsilon - \gamma_c}\frac{\tau^y}{1 - \tau^y} + \eta \mathcal{I}_y^S + \zeta_t \mathcal{I}_c^S\right)\frac{U_S}{U_c}\right]\end{aligned} \tag{25}$$

B.2 Savings multiplier over the life cycle

Fix a realized history. Let $\kappa_t = (1 - \bar{\tau}_t^s)\beta R^2 > 0$, $P_t = \prod_{j<t} \kappa_j$, with $P_0 = 1$, and $x_t = \vartheta_t / P_t$. Equation (12) gives

$$x_{t+1} = x_t - \frac{\eta_t^s}{P_t U_{c,t}} \qquad t < T. \tag{26}$$

The endpoint conditions $\vartheta_0 = \vartheta_{T+1} = 0$ imply $x_0 = 0$ and $x_T = \eta_T^s / (P_T U_{c,T})$. Summing (26),

$$\sum_{t=0}^{T} \frac{\eta_t^s}{P_t U_{c,t}} = 0$$

If any multiplier along the history is nonzero, terms of both signs must appear. We have

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R}\mathbb{E}_t\left[\frac{1 - \chi_{t+1}}{U_{c,t+1}}\right] \tag{27}$$

If we define $\mathcal{G}_\sqcup \equiv 1 - \gamma_{c,t}\frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Gamma_{c,t}$,

$$1 - \chi_t = \mathcal{G}_t - \eta_t^s \frac{\sigma_t}{c_t} \tag{28}$$

Where $\mathcal{G}_t > 0$ and $\mathbb{E}_t[\mathcal{G}_{t+1}] > 0$, define the reference wedge by $1 - \tau_t^{s,*} = \mathcal{G}_t / \mathbb{E}_t[\mathcal{G}_{t+1}]$. Multiplying (27) by $U_{c,t}$ and substituting the Euler equation with the fixed savings wedge gives

$$1 - \chi_t = (1 - \bar{\tau}_t^s)\left\{\mathbb{E}_t[1 - \chi_{t+1}] - \mathrm{Cov}_t\left(U_{c,t+1}, \frac{1 - \chi_{t+1}}{U_{c,t+1}}\right)\right\}$$

Substitution of (28) then yields

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s)\, \mathbb{E}_t \left[\eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} \right] = \mathbb{E}_t[\mathcal{G}_{t+1}](\bar{\tau}_t^s - \tau_t^{s,*}) + (1 - \bar{\tau}_t^s)\, \mathrm{Cov}_t \left(U_{c,t+1}, \frac{1 - \chi_{t+1}}{U_{c,t+1}} \right)$$

When the next draw is degenerate this reduces to

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} = \mathcal{G}_{t+1}(\bar{\tau}_t^s - \tau_t^{s,*}) \tag{29}$$

For a deterministic history, let $z_t = \eta_t^s \sigma_t / c_t$ and assume $\sigma_t > 0$, $\mathcal{G}_t > 0$, and $1 - \bar{\tau}_t^s > 0$. Strict curvature is needed because Assumption 2.2 alone allows $\sigma_t = 0$. Equation (29) reads

$$z_t = 1 - \bar{\tau}_t^s z_{t+1} \mathcal{G}_{t+1}(\bar{\tau}_t^s - \tau_t^{s,*})$$

If every $g_t > 0$, then $z_t \leq 0$ implies that $z_{t+1} < 0$. We know some multipliers mut be nonzero, so the sequence starts positive and ends negative, crossing once. If every $g_t < 0$, this reverses.

C Derivations: linear carbon price

C.1 Expenditure, demand, and incentive compatibility

To simplify notation, let $\bar{p}_t \equiv 1 + \bar{\tau}_t^q$. The planner has the below dirty good wedge constraint:

$$\Gamma_t^q \equiv \frac{U_{q,t}}{U_{c,t}} - \bar{p}_t = 0$$

At fixed price, labor, and pollution, define indirect utility by

$$\tilde{U}(x, l; \bar{p}, S) = \max_{c,q} \; U(c, q, l, S) \quad \text{s.t.} \quad c + \bar{p}q = x$$

Here, x is the total expenditure on the two goods. If we substitute $c = x - \bar{p}q$, we get the first-order condition $U_q = \bar{p}U_c$. With a linear dirty good wedge, the planner is essentially only choosing each household's total expenditure x and labor l rather than also the consumption mix, which is now equal for everyone. If we write $\tilde{U}(x, y/\theta; \bar{p}_t, S_t)$, we have $\tilde{U}_x = U_c$ because we can express x as a function of c. Now that the planner cannot impose a consumption mix on households, it chooses the linear dirty good wedge and total expenditure such that incentive constraints are satisfied.

$$\tilde{U}\left(x(\hat{\theta}), \frac{y(\hat{\theta})}{\theta}; \bar{p}_t, S_t\right) \geq U\left(c(\hat{\theta}), q(\hat{\theta}), \frac{y(\hat{\theta})}{\theta}, S_t\right)$$

Differentiate $U_q = \bar{p}U_c$ jointly with the expenditure constraint. Define $\mathcal{D} \equiv U_{qq} - 2\bar{p}U_{cq} + \bar{p}^2 U_{cc}$. At fixed labor and pollution, expenditure and compensated price changes give

$$\begin{aligned} \frac{\partial q^d}{\partial x} &= \frac{U_{cq} - \bar{p}U_{cc}}{|\mathcal{D}|} = \frac{\bar{p}U_c \mathcal{I}_c^q}{|\mathcal{D}|} \\ s_{qq} \equiv \left.\frac{\partial q^d}{\partial \bar{p}}\right|_u &= \frac{U_c}{\mathcal{D}} < 0 \end{aligned}$$

From here, we can write

$$\Delta_t^q \equiv \bar{p}\mathcal{I}_c^q - \mathcal{I}_q^q = -\frac{\mathcal{D}}{\bar{p}U_c} = \frac{1}{\bar{p}|s_{qq}|} > 0 \tag{30}$$

C.2 Optimal linear dirty good wedge

The Lagrangian of Appendix A.1 gains $\int \eta^q \Gamma^q dF^t$. Using the usual willingness-to-pay identities,

$$\Gamma^q_z = \frac{U_{qz} - \bar{p} U_{cz}}{U_c} = \bar{p}\, \mathcal{I}^q_z. \tag{31}$$

Each pointwise condition for c, q, and l gains $\eta^q f \Gamma^q_z$. With income wedges free,

$$\frac{\phi U_c}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta^q \bar{p}\, \mathcal{I}^q_c \tag{32}$$

$$\frac{\phi U_c}{f} = \frac{1 + \nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} + \eta^q \mathcal{I}^q_q \tag{33}$$

Here ν_t is the shadow emission price in (15). Equating (32) and (33) gives

$$\underbrace{1 - \frac{1 + \nu_t}{\bar{p}_t}}_{\varpi_t} = \underbrace{(\gamma_c - \gamma_q) \frac{\psi U_c}{\theta f}}_{\varpi^*_t} - \eta^q_t \Delta^q_t$$

Using (30),

$$\eta^q_t = \bar{p}_t |s_{qq,t}| (\varpi^*_t - \varpi_t)$$

The common wedge enters the date-0 resource-cost Lagrangian only through the date-t price constraints. Accounting for resource discounting and transition probabilities gives

$$\frac{d\mathcal{L}_0}{d\bar{\tau}^q_t} = -\frac{1}{R^t}\, \mathbb{E}_0[\eta^q_t]$$

The planner minimizes resource cost at fixed promised welfare, so a negative derivative favors raising the dirty good wedge. An interior optimum sets this derivative to zero. Since $\bar{p}_t$ and ϖ_t are common across date-t histories,

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}| \varpi^*_t]}{\mathbb{E}_0[|s_{qq,t}|]}$$

Weak separability gives $\varpi^*_t = 0$ at every history, so the optimal date-specific rate is ν_t.

C.3 Fixed income wedges and a lifetime rate

The conditions for c, q, and l gain $\eta^q f \Gamma_z^q$. Equating the resulting conditions for c and q,

$$\varpi_t = \varpi_t^* - \eta_t \left(\Gamma_c - \frac{\Gamma_q}{\bar{p}_t} \right) + \eta_t^s \mathcal{I}_{c,t}^q - \eta_t^q \Delta_t^q$$

Using the Γ definitions and rewriting as in Appendix B.1,

$$\Gamma_c - \frac{\Gamma_q}{\bar{p}_t} = (1 - \bar{\tau}_t^y) \mathcal{I}_{c,t}^q \frac{\gamma_{c,t} - \gamma_{q,t}}{y_t}$$

Putting the $\mathcal{I}_{c,t}^q$ terms together,

$$\varpi_t = \varpi_t^* + \eta_t \frac{\gamma_{c,t} - \gamma_{q,t}}{y_t} - ((1 - \bar{\tau}_t^y)\eta_t - \eta_t^s) \mathcal{I}_{c,t}^q - \eta_t^q \Delta_t^q$$

Alternatively,

$$\varpi_t = \varpi_t^* + \eta_t \frac{\gamma_{c,t} - \gamma_{q,t}}{y_t} - \zeta_t \mathcal{I}_{c,t}^q - \eta_t^q \Delta_t^q$$

With labor and savings wedges fixed, the linear dirty good wedge satisfies

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}| \ \varpi_t^{**}]}{\mathbb{E}_0[|s_{qq,t}|]}$$

If the planner must not only hold the dirty good wedge constant across households in a given date but also across dates, summing across dates and substituting the multiplier formula,

$$\sum_{t=0}^{T} \frac{1}{R^t} \mathbb{E}_0[|s_{qq,t}|(\varpi_t^{**} - \varpi_t)] = 0$$

For a time-varying social cost of carbon, $\varpi_t = (\bar{\tau}^q - \nu_t)/\bar{p}$ and this is equivalent to

$$\bar{\tau}^q = \frac{\sum_{t=0}^{T} \frac{1}{R^t} \mathbb{E}_0[|s_{qq,t}|(\nu_t + \bar{p}\varpi_t^{**})]}{\sum_{t=0}^{T} \frac{1}{R^t} \mathbb{E}_0[|s_{qq,t}|]}$$

C.4 Social cost of carbon

The linearity constraint affects the damage weight through both the direct pollution derivative and the delivery cost ϕ. From (31) and (32), their combined contribution at each history is

$$\eta^q \left(\Gamma_S^q - \Gamma_c^q \frac{U_S}{U_c} \right) = -\eta^q \frac{U_S}{U_c} \left(\bar{p} \mathcal{I}_c^S - \mathcal{I}_q^S \right)$$

This follows directly from $\mathcal{I}_z^S = \frac{\partial \ln(-U_S/U_c)}{\partial z}$ and the quotient rule. With income wedges free, substituting into the condition of Appendix A.5 yields

$$\mu_t - \frac{1}{R} \frac{\partial V_{t+1}}{\partial S_t} = -\mathbb{E}_0 \left[\left(1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta^q \left(\bar{p} \mathcal{I}_c^S - \mathcal{I}_q^S \right) \right) \frac{U_S}{U_c} \right]$$

The expectation averages all date-t histories. With income wedges fixed, this adds the same η^q term to the bracket of (25). The emission recursion (15) is unchanged, so aggregating the current-date costs on the right-hand side as in Appendix A.5 gives the social cost of carbon.

Choosing the common wedge optimally does not generally remove the additional term. At a date whose price is free to adjust, $\mathbb{E}_0[\eta_t^q] = 0$ and its contribution to the social cost of carbon is

$$\mathbb{E}_0[\eta_t^q \mathcal{K}_t] = \mathrm{Cov}_0(\eta_t^q, \mathcal{K}_t) \qquad \mathcal{K}_t \equiv -\frac{U_{S,t}}{U_{c,t}} \left(\bar{p}_t \mathcal{I}_{c,t}^S - \mathcal{I}_{q,t}^S \right)$$

The social cost remains common across households because pollution is aggregate, but its level can change with the constrained allocation.